IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 18 May 2025, accepted 16 June 2025, date of publication 24 June 2025, date of current version 2 July 2025.

*Digital Object Identifier 10.1109/ACCESS.2025.3582318*

RESEARCH ARTICLE

# A Normalized Exponential Piecewise Chaotic System (NEPCS) and DNA Image Cryptography Using SHA-256

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This work was supported in part by the VTT-Technical Research Centre of Finland Ltd.

**ABSTRACT** One-dimensional (1D) chaotic systems are crucial in cryptography to encrypt digital images effectively by transforming them into noise. Despite their simplicity and computational efficiency, 1D chaotic systems often struggle with short periodicity, narrow control parameter ranges, less complex phase space attractors, and regularity in their output, which limit their applicability in cryptography. A novel 1D piecewise chaotic map, called Normalized Exponential Piecewise Chaotic System (NEPCS), is proposed with an extensive range of control parameters $\lambda$ and prime number $p$. It has a high score for Lyapunov exponent ($> 7.88$), sample entropy score of 2.2137 and correlation dimension score exceeding 1.1. The NEPCS is evaluated using the SP800-22 test suite recommended by the National Institute of Standards and Technology (NIST). The qualifying rate for the NIST test is $> 0.49$, proving its high quality of randomness. To validate the efficacy of the NEPCS, an image encryption technique is introduced, leveraging the SHA-256 hash function used iteratively on keyboard inputs and plain images to derive initial seeds for NEPCS. The encryption process begins by permuting the pixels of a color image and converting them into DNA bases. The cipher block chaining mode (CBC) of substitution is sandwiched between the keyless transpositions. The exclusive-or and addition operations are alternately applied to substitute the DNA bases. The presented image cipher has NPCR $> 99.59\%$, UACI exceeding 33.40%, $\chi^2 < 285.60$, and entropy $\approx 7.9999$ scores while resisting communication noises.

**INDEX TERMS** 1D chaos, normalized exponential piecewise chaotic system (NEPCS), DNA, image encryption, CBC.

## I. INTRODUCTION

With the advancement in internet technology, the multimedia information that contains images, videos, and audio is frequently shared among people. The use of images in data sharing has been steadily increasing, as they represent one of the most comprehensive forms of information, capable of overcoming language barriers and enhancing communication. The extensive use of images over social networks has led to security issues, particularly in fields like the military, business domains and medical organizations [1], [2], [3]. The unauthorized access to sensitive and private information may cause privacy leakage, raising the need for the appropriate security measures [4], [5], [6]. Therefore, image encryption can be applied to secure the data and control the access to the sensitive information [7], [8], [9].

The associate editor coordinating the review of this manuscript and approving it for publication was Mohamad Afendee Mohamed.

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In the past years, researchers have introduced chaos-based encryption techniques [10], [11], [12]. The standard text encryption algorithms like AES (Advance Encryption Standard) and DES (Data Encryption Standard are unsuitable for digital image encryption due to redundancy, strong correlation among the pixels and bulky size [4], [13], [14]. An image encryption method should be more effective and secure to ensure the secrecy and integrity of the information. Numerous image encryption methods, including chaos-based encryption techniques, were introduced to improve the security [15], [16], [17]. The chaotic systems and encryption techniques share properties like sensitivity to the initial conditions, randomness, and determinism that make chaos suitable for designing encryption algorithms. The ergodicity, determinism, and pseudo-randomness are beneficial to make the image encryption techniques more robust and secure [18], [19], [20], [21].

In chaos-based encryption techniques, digital images are encrypted with random sequences generated by the non-linear system by passing the secret key. According to Shannon's information theory, permutation and diffusion are two important and basic processes in developing image encryption techniques [22], [23], [24]. In the permutation phase, the position of the pixels is altered while the intensity of the image remains the same, this reduces the correlation among the pixels [23], [25], [26], [27], [28]. In the diffusion phase, the pixel values are modified by manipulating them with other values. The security of the permuted image can be increased by using the diffusion process [29], [30], [31].

### A. RELATED LITERATURE

The encryption incorporates changing the pixel's position and manipulating the pixel value in its most basic form. In image encryption, random numbers generated by non-linear dynamical systems are commonly used to permute and diffuse the pixels. Sun and Wang utilized a novel hyperchaotic system to generate the chaotic sequences, SHA-512 for creating the secure hash key, and DNA encoding operations for the image encryption [32]. Soni et al. devised a new 1D chaotic map called M-Map and presented a comparative analysis with the 1D chaotic maps against their dynamical properties. Soni et al. aimed to visualize the suitability of the chaotic maps to build lightweight encryption systems for smart grid environments. Further, a combination of the chaotic maps was used to prepare a potentially secure crypto-system to address the data security challenges [33]. Ullah et al. introduced a Cosine chaotic Map and analyzed its robustness in color image encryption. The encryption system works in three stages: creating circular wheels from color channels, rotating wheels around angle theta for pixel permutation and wheels exclusive or with each other in cipher block chaining mode for substitution. The angle theta is derived from the Cosine chaotic system [4].

The classical Logistic map was modified by Alawida et al. with the help of the anterior perturbation to enhance the randomness and chaotic range of the map. The modified Logistic map, Enhanced Logistic Map termed as EL Map, exhibited an improved chaotic behaviour over a large chaotic range compared to the original map [34]. Authors Huang et al. have employed a crossbreeding fusion of the pseudo-random number generators and genetic operations for the digital image encryption [35]. A novel chaotic map was introduced and rigorously analyzed by Rongjun Ge et al. that was a good source of pseudo-random numbers [36]. Nazish et al. introduced a new non-linear dynamical system based on the Fibonacci sequence, with enhanced chaotic behaviour in the domain of one-dimensional maps has been introduced [37]. A novel image encryption algorithm is proposed by Bouteghrine et al. with a newly designed 3D chaotic map. The algorithm employed both pixel scrambling and diffusion, leading to a more secure encryption method with the large key space than the existing methods [38]. Mou et al. utilized multiple chaotic systems to develop the chaos-based encryption algorithm and also introduced the improved version of the Joseph block scrambling technique to enhance the confusion and diffusion process [39].

Although 1D chaotic maps are used in a large variety of cryptographic applications due to their simple and cost-effective nature, they possess a short chaotic range and exhibit weak chaos, with a correlation dimension score < 1 [40] and a Sample entropy score < 1.5 [41]. The classical Logistic map has the Sample entropy $SE \approx 0.7$ [11] and $D_2 \approx 0.99$ [42]. The short chaotic range of the Logistic map with the highest Lyapunov exponent $\lambda$ is 0.6724 [42], and they structured phase space attractor denotes its predictability and low randomness. These shortcomings of 1D chaotic maps make it less reliable to design a secure encryption system [33].

To address the above limitations of the 1D chaotic system, a novel 1D piece-wise chaotic map, termed as **Normalized Exponential Piece-wise Chaotic System (NEPCS)**, is introduced. The NEPCS features deterministic chaotic behavior over an exceptionally large range of the control parameters, exhibiting high ergodicity and extreme sensitivity to the minute changes in the control parameters. The positive Lyapunov exponent and impressive sample entropy score prove its robustness. It has been validated using the recommended NIST test suit SP800-22 and widely adopted TestU01. The NEPCS demonstrated its potential as a secure and reliable addition to the domain of 1D chaotic systems for designing a more robust image encryption scheme. The key contributions of the work for the proposed chaotic system are described as follows:

- A novel chaotic system called Normalized Exponential Piecewise Chaotic System (NEPCS) is introduced to generate deterministic pseudo-random numbers.
- The proposed NEPCS has two control parameters $\lambda$ and prime number $p$, which substantially enhances its key space.
- The NEPCS will be evaluated through Lyapunov Exponent, Bifurcation Diagram, Phase Space attractor,

Correlation Dimension $D_2$, Sample Entropy, Uniform Distribution analyses, etc., and NEPCS will be compared with some of the 1D chaotic systems.
- The NEPCS will be subjected to the NIST SP-800-22 test suite and TESTU01 to validate its applicability in cryptographic applications.

An image encryption technique has been developed to assess the NEPCS's efficacy further. The proposed image encryption technique achieves high NPCR, UACI, Chi-Square, and entropy scores while demonstrating a strong resistance to communication noise. Here are some key contribution points of the proposed image encryption algorithm;

- The SHA-256 is applied iteratively on keyboard inputs combined with the hash of the plain image, to generate seeds for the NEPCS system. This ensures the seeds depend on the plain image to avoid chosen plaintext attacks.
- The pixels are encoded into DNA bases using four out of eight DNA encoding rules, selected randomly for every pixel.
- The keyed permutation is applied on the pixels of the plain image, and then chaotically selected DNA rules are used to encode.
- The keyless permutation known as zigzag is applied to perform substitution at a 2-bit level.
- Cipher Block Chaining (CBC) substitution mode is performed on DNA bases using dual operations, addition and exclusive-or applied in an alternating fashion.
- Another 2-bit level substitution by permutation is applied called snake pattern.
- One of the eight DNA rules is selected for decoding, chosen randomly via computed value from the hash of the SHA-256 function.

The research article is divided into six sections. Section II discusses the related work, and section III describes the proposed non-linear system, the NEPCS, and its performance comparisons. Section IV reviews the rules and operations of Deoxyribonucleic Acid (DNA). Section V provides a detailed explanation of the image encryption algorithm, followed by results and comparisons. The final section concludes the study.

## II. PRELIMINARY IDEAS

### A. LOGISTIC MAP

The Logistic map is a classical chaotic map introduced by May in 1976 [43]. Its mathematical structure is given as:

$$x_{n+1} = \mu x_n(1 - x_n) \tag{1}$$

In the above Equation (1), the $\mu$ is the control parameter in the range [0, 4]. The logistic map is chaotic for $3.56995 > \mu > 4$ and shows periodic windows at $3.85724 > \mu > 3.82843$ as shown in Figure 1(a) to prove the stated behavior of the Logistic map.

### B. ENHANCED LOGISTIC MAP

The Enhanced Logistic (EL) map defined by Alawida [34] in 2024 and its mathematical description is given as follows:

$$x_{n+1} = \frac{F_L + 1}{x_n} \mod 1 \tag{2}$$

As defined in Equation (2), the EL map is a variant of the classical Logistic map, $F_L$. It demonstrates chaotic dynamics when the parameter $4 > \mu > 0$. This chaotic behavior is confirmed by the bifurcation diagram in Figure 1(b) which clearly illustrates the EL map's chaotic nature within this range of $\mu$.

### C. PWLCM

The Piece-Wise Linear Chaotic Map (PWLCM) introduced by Zhou et al. [44] is mathematically described as:

$$x_{n+1} = \begin{cases} \dfrac{x_n}{\gamma} & 0 \le x_n < \gamma \\ \dfrac{x_n - \gamma}{0.5 - \gamma} & \gamma \le x_n < 0.5 \\ F(1 - x_n) & 0.5 \le x_n < 1 \end{cases} \tag{3}$$

In Equation (3), $\gamma$ is the control parameter, which lies in the range [0, 0.5]. The system is chaotic and generates pseudo-random numbers within the range [0, 1], and this fact is presented in Figure 1(c), where the bifurcation diagram is displayed.

## III. NORMALIZED EXPONENTIAL PIECEWISE CHAOTIC SYSTEM (NEPCS)

A novel 1D pseudo-random number generator with a substantial key space is proposed as a Normalised Exponential Piece-Wise Chaotic System (NEPCS), mathematically described as:

$$x_{n+1} = \begin{cases} \lambda \times e^{-x_n^2} \times p \times x_n & mod 1 \quad if\, x_n < 0.5 \\ -\lambda \times e^{x_n^3} \times x_n & mod 1 \quad if\, x_n \ge 0.5 \end{cases} \tag{4}$$

where $\lambda$ serves as the control parameter and $x_0$ represents the initial seed in the range of [0 - 1] with the precision of $10^{15}$. The *mod* function is utilized to constrain the random values between 0 and 1. For better chaotic behavior and to deceive the attackers, initial conditions provided to the chaotic system must not be integer values such as 1, 2, or 3. The use of **exp** function and two parameters, the prime number $p$ and $\lambda$ complicate its structure and behavior to become unpredictable.

### A. PERFORMANCE ANALYSIS

The assessment of the chaotic performance of the proposed NEPCS is essential to identify its chaotic features. In this section, various metrics are used to evaluate the output trajectory, key sensitivity, range of control parameter for which a non-linear system exhibits chaotic behavior, and the quality of the pseudo-randomness of NEPCS. For comparative study, the Logistic, EL map, and PWLCM

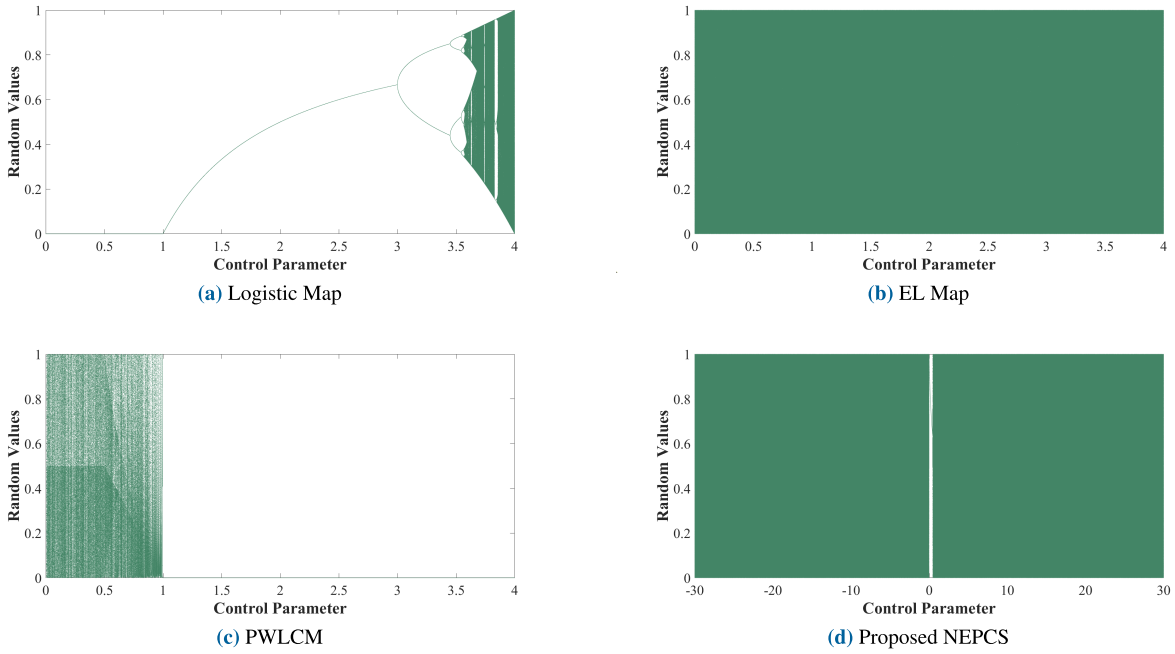



**FIGURE 1.** Bifurcation analysis of four chaotic systems.

chaotic maps are selected and preliminaries of these systems are already discussed in section II.

The fixed value of initial seed $x_0 = 0.0000000001$ and control parameters $\mu = 4$ (Logistic), $\mu = 3.99$ (Enhanced Logistic), $\gamma = 0.49$ (PWLCM), $\lambda = 29.99$ (NEPCS), prime number $p = 97$ (NEPCS) are used for all four systems to generate random numbers for bifurcation, Lyapunov Exponent, phase space, key sensitivity and histogram analysis. The size of pseudo random vectors is different for each test depending upon the requirement.

### 1) TRAJECTORY ANALYSIS

The trajectory analysis is crucial in understanding the chaotic behavior of a non-linear system as it depicts the trajectory of generated random numbers. This analysis evaluates the area covered by the trajectory and the distance between the two trajectories [45]. A robust chaotic system should feature trajectories that diverge exponentially and do not closely overlap. To perform this analysis, the bifurcation and 3D phase planes are plotted for the proposed map alongside baseline chaotic maps.

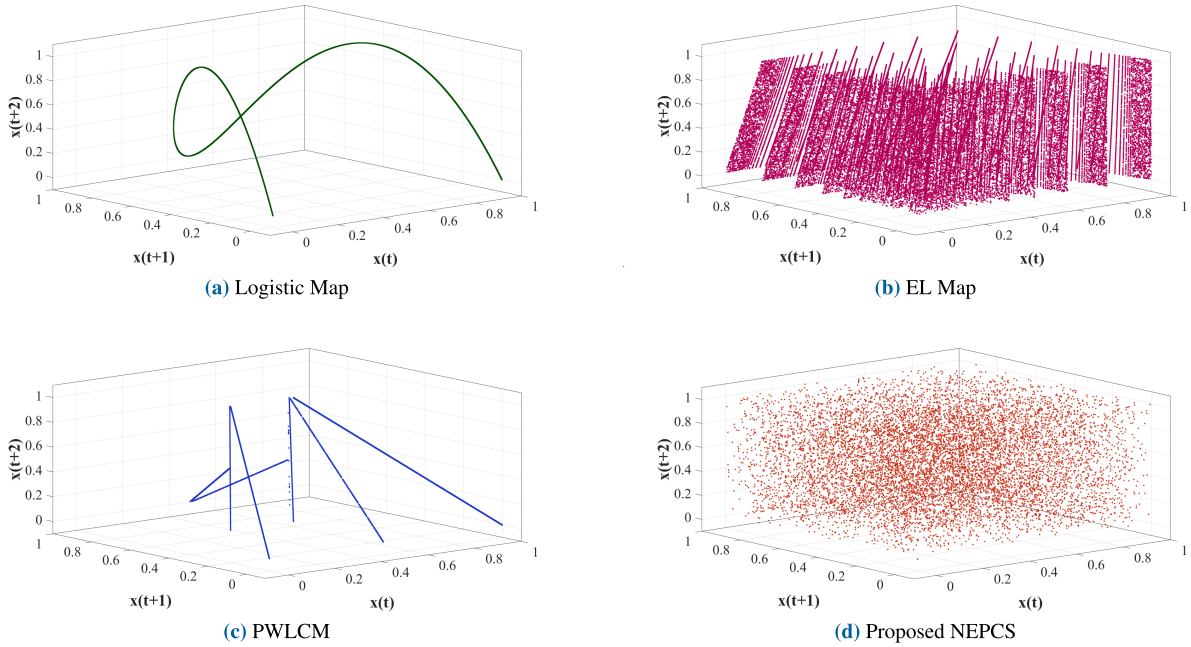
Figure 1 shows the trajectories of all the chaotic systems as a function of their control parameters. It is evident that the Logistic map and PWLCM exhibit relatively short ranges, with their trajectories observable only for $(3.82842 < \mu < 4)$ and $(0 < \gamma < 0.5)$, respectively. In contrast, the Enhanced Logistic map shows a slightly broader range for $(0 < \mu < 4)$. The proposed chaotic system has tested within the range of $(-30 < \lambda < 30)$ excluding the range $[-0.5$ to $+0.5]$ for which it shows non-chaotic behavior. The precision of $\lambda$ is $10^{-12}$. This proves that the system exhibits chaotic behavior across a wider range of dynamics and has the potential for greater complexity and versatility.

The second analysis involves the construction of 3D phase plane of the Logistic map ($\mu = 4.0$), Enhanced Logistic map ($\mu = 4.0$), PWLCM ($\gamma = 0.123$), and the proposed NEPCS ($\lambda = 2.914$) and are depicted in Figure 2. It can be observed that the Logistic map and PWLCM exhibit closed trajectories within a narrow range of control parameters shown in Figure 2(a) and 2(c) while the EL map illustrates more distant trajectories across a wider range compared to the Logistic map and PWLCM as in Figure 2(b). However, it still does not perform well as the proposed NEPCS, which displays a highly expansive chaotic range for $(-30 < \lambda < 30)$ depicted in Figure 2(d).

### 2) SENSITIVITY ANALYSIS

The sensitivity of chaotic systems to the initial conditions is a key factor in determining their robustness. A chaotic system should be capable of generating entirely distinct sequences from different initial seeds, even if the seeds differ by just a single bit.

The Lyapunov exponent serves as a crucial measure of this sensitivity, quantifying how the system's behavior responds to changes in its key parameters. It assesses the divergence or convergence of two trajectories originating from nearby initial conditions [46]. A positive Lyapunov exponent indicates that the trajectories diverge exponentially over time, which is a hallmark of chaotic behavior in the system. By examining the Lyapunov exponent spectrum, one can gain insights into the underlying dynamics and identify the presence of chaos within the system. The mathematical




(a) Logistic Map (b) EL Map

(c) PWLCM (d) Proposed NEPCS

**FIGURE 2.** 3D Phase space analysis.

definition of the Lyapunov Exponent is as follows:

$$LE = \lim_{x\to\infty}\left(\frac{1}{n}\sum_{i=0}^{n-1} In\left|g'(x_i)\right|\right) \tag{5}$$

In Equation (5), $g'(x_i)$ is the derivative of the function at a particular point $x_i$. Figure 3(a) illustrates the Lyapunov exponent spectrum for the Logistic map, EL map, PWLCM, and NEPCS as a function of their respective parameters. Evidently, the maximum Lyapunov exponent of NEPCS greatly exceeds the largest Lyapunov exponents of the Logistic map, EL map, and PWLCM. Additionally, the distribution of positive Lyapunov exponents highlights the extensive chaotic range of NEPCS, as illustrated in Figure 3(b).

To further assess the key sensitivity of NEPCS, the analysis of two trajectories $S_1$ and $S_2$ are generated using seeds $x_1 = 0.00000000001$ and $x_2 = 0.00000000002$ while the control parameters remain the same as provided in the III-A for all of the selected chaotic systems. The first 40 iterations of these sequences are plotted in Figure 4 for the four chaotic systems. The sequences generated via NEPCS begin to diverge from the $4^{th}$ iteration, while this divergence started at $15^{th}$ iteration for Logistic, $6^{th}$ for EL map and $26^{th}$ for PWLCM. This divergence demonstrates that the NEPCS is extremely sensitive to initial conditions.

Similarly, the NEPCS sensitivity is also analyzed with slight variations in control parameters $\lambda$ and $p$, while fixing the initial seed $x_1 = 0.0000000001$. The small differences in $\lambda$ switching from 29.99 to 29.98 break the chaotic pattern as illustrated in Figure 5(a) at the $5^{th}$ iteration. When prime number $p$ changes from 97 to 101, the NEPCS proved its sensitivity as both sequences started diverging at $4^{th}$ iteration depicted in Figure 5(b). The proposed NEPCS exhibited a sensitive response toward varying control parameters and seed values, demonstrating higher effectiveness for security applications.

### 3) HISTOGRAM

The histogram of the time series represents the output distribution of the chaotic map, the more uniform the distribution, the better the quality of the map's output. It represents the output space of a chaotic system by dividing it into subintervals, allowing for the analysis of the progression of the system within those intervals. The height of each subinterval in the histogram reflects the frequency density within that interval. For $n$ number of random numbers, the number of subintervals $m$ is determined via square root rule $m = \sqrt{n}$ [47]. To perform this measure, a total $1 \times 10^6$ observations against each chaotic map is generated yielding a histogram with $m = 1000$ bins. The reader can view in Figure 6, the histogram of NEPCS along with the baseline chaotic systems. It is evident that the Logistic map, EL map, and PWLCM exhibit non-uniform distributions, while the NEPCS demonstrates a uniform distribution across the subintervals without exhibiting any periodic, upward, or downward pattern. Outliers, characteristic of problematic non-linear systems, are observed in the baseline chaotic maps. In contrast, the proposed system exhibits unstable and highly random behavior, making it particularly suitable for cryptographic applications.

### 4) CORRELATION DIMENSION

The correlation dimension $D_2$ is a quantitative measure of the geometric complexity of an attractor in a non-linear dynamical system. It quantifies the distance relationship among

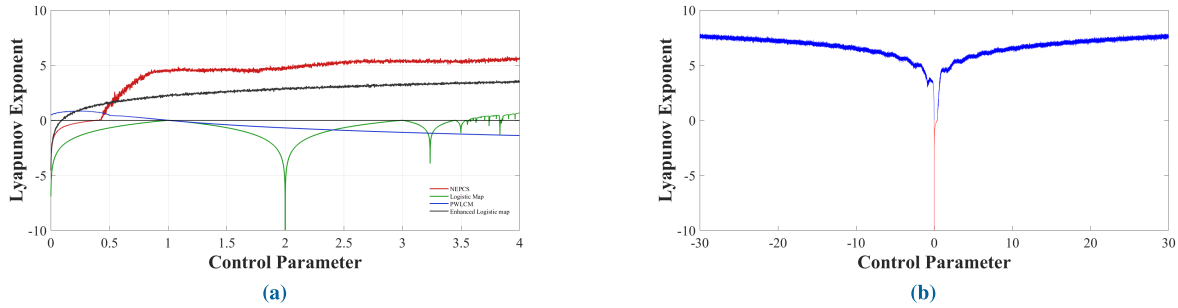



**FIGURE 3.** Analysis of Lyapunov Exponent: (a) Logistic Map, EL Map, PWLCM and NEPCS (b) NEPCS Only.

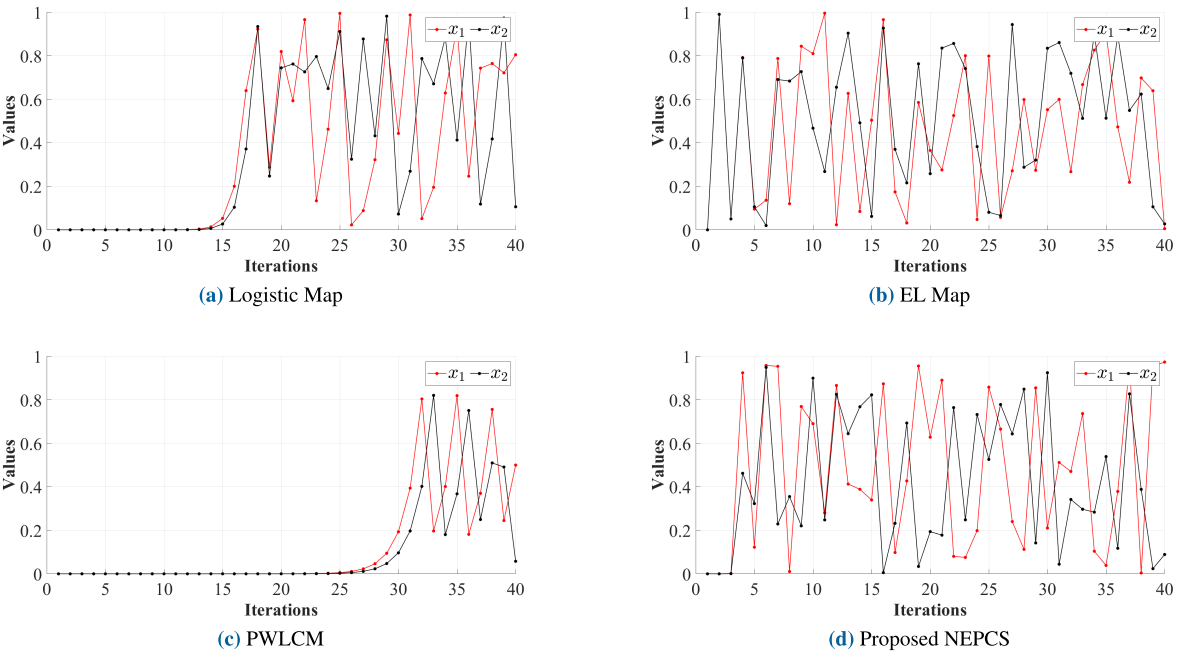



**FIGURE 4.** Initial seed sensitivity analysis.

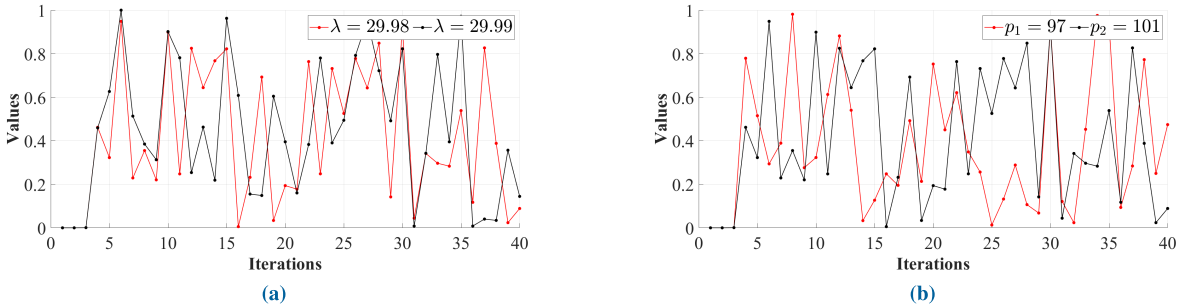
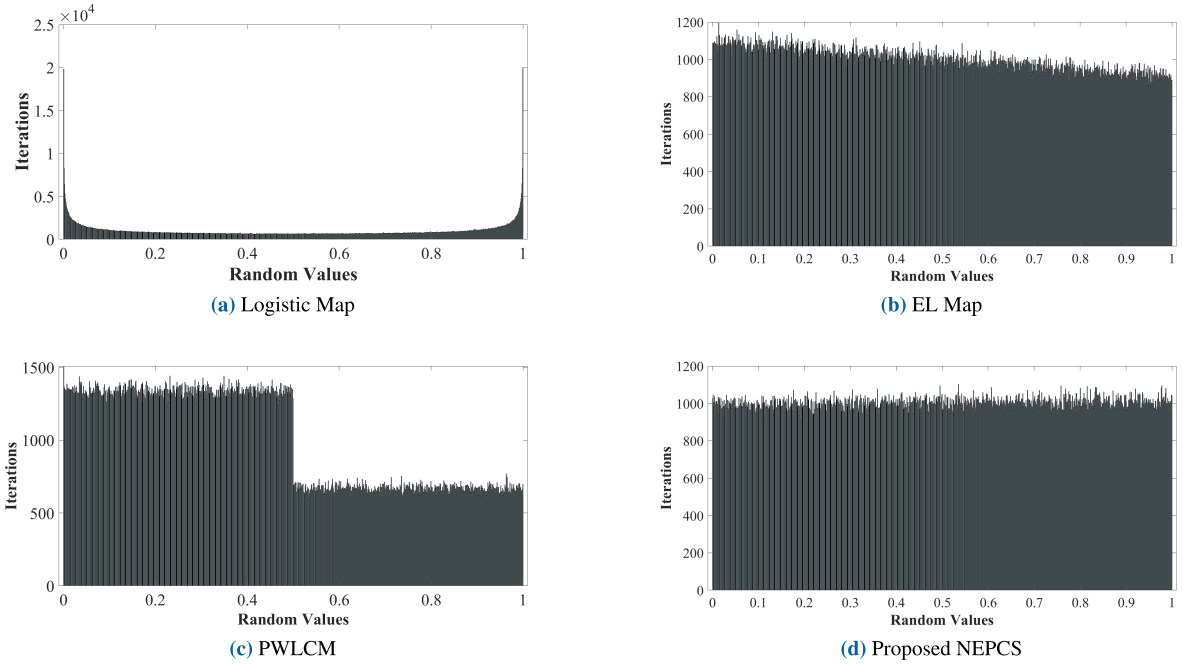



**FIGURE 5.** Sensitivity analysis of control parameters of Proposed NEPCS (a) Lambda $\lambda$ (b) prime number $p$.

the points of a phase space to distinguish between random noise and deterministic chaos. The Grassberger-Procaccia algorithm (1983) is commonly used to estimate the correlation dimension [40], [48]. The $D_2$ is computed for the phase space trajectories $y_n = \left(x_n, x_{n+\tau}, x_{n+2\tau}, \ldots, x_{n+(m-1)\tau}\right)$ constructed from the random sequence $x(n)$ with embedding dimension $m$ and the lag value $\tau$. The mathematical expression of $D_2$ is given in Equation (6), calculated using the correlation integral in Equation (7).

$$D_2 = \lim_{r \to 0} \frac{\ln C(r)}{\ln r}. \tag{6}$$

$$C(r) = \frac{1}{N(N-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^{N} H(r - d(y_i - y_j)). \tag{7}$$




(a) Logistic Map (b) EL Map (c) PWLCM (d) Proposed NEPCS

**FIGURE 6.** Comparative analysis of Histogram for all four chaotic system.

and

$$H(x) = \begin{cases} 1, & \text{if } x \geq 0, \\ 0, & \text{if } x < 0. \end{cases} \tag{8}$$

$$d(y_i, y_j) = \sqrt{\sum_{k=1}^{m} (y_{ik} - y_{jk})^2}. \tag{9}$$

where $y_{ik}$ and $y_{jk}$ are two data points $y_i$ and $y_j$ in the $k$-th dimension.

The $D_2$ is computed under different conditions to analyze the chaotic behavior of the proposed system compared to baseline chaotic maps. The analysis has been conducted with a minimum value of the initial seed $x_0 = 0.0000000001$ along with 100 randomly selected values of the control parameter for each system within its defined range and the results are displayed in Figure 7(a). The minimum value of $D_2$ for NEPCS is 1.0263 and max value is 1.3984 which is higher than the minimum score of 0.9974 and the maximum value of 1.0261 for EL Map. In another experiment, an additional 100 values of the correlation dimension $D_2$ have been computed using a fixed seed value of $x_0 = 0.9999999999$ while the control parameters were selected randomly for each system depicted in Figure 7(b). The $D_2$ values for the NEPCS are close to 1.5, while the other maps exhibit lower values. This indicates that the NEPCS has the most complex attractor among the four systems.

The $D_2$ values represented in Figure 8 are computed for a fixed control parameter as discussed in the section III-A along with 100 randomly selected values of the initial seed within range [0-1]. The NEPCS has $D_2$ value in the range of 1.31799 to 1.42401 while all other systems have far lower $D_2$ scores depicted in Figure 8. Finally, the $D_2$ scores are computed using fixed seed value and control parameters for all four chaotic systems and results are given in Table 1. The results demonstrate that the proposed system exhibits more complex and unpredictable behavior than existing chaotic maps as it has the highest $D_2$ value. This chaotic behavior strengthens the security of NEPCS and making it highly effective for practical cryptographic applications.

**TABLE 1.** $D_2$ results for NEPCS ($\lambda = 29.99$), Logistic Map ($\mu = 4$), EL Map ($\mu = 3.99$), and PWLCM ($\gamma = 0.49$) and initial Seed = $x_0 = 0.0000000001$.

| PRNG | $D_2$ |
|---|---|
| NEPCS | 1.4877 |
| Logistic Map | 0.9054 |
| Enhanced Logistic Map | 1.0071 |
| PWLCM | 0.9941 |

### 5) SAMPLE ENTROPY

Sample entropy is another statistical measure that quantifies the randomness and unpredictability of a dynamic system. It can be calculated using Equation 10.

$$SE(m, r, N) = -\log \frac{A}{B} \tag{10}$$

where $A$ and $B$ are two random sequences defined via: $d(X(i+1), X(j+1)) \leq r$ and $d(X(i), X(j)) \leq r$ respectively, and $d(X(i), X(j))$ is distance between two random data points $X(i)$ and $X(i+1)$. The $m = 3$ is the embedding dimension and $r = 0.2 \times std$ is the threshold value, and $std$ is the standard deviation of a random sequence. For a non-linear system to

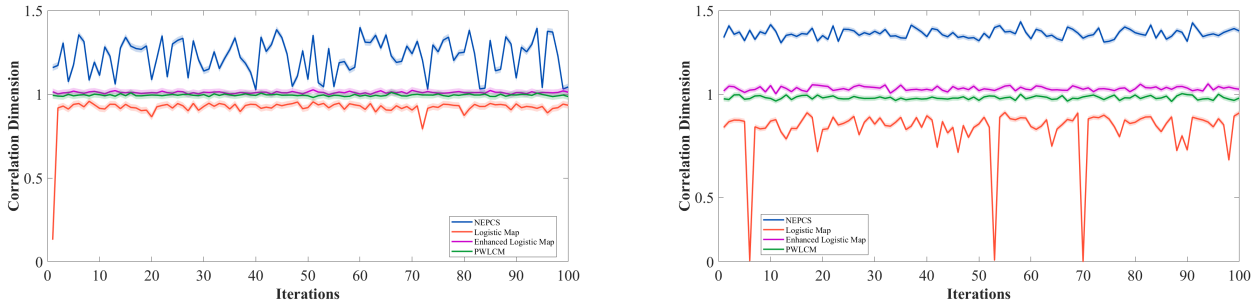



**FIGURE 7.** $D_2$ results for NEPCS, Logistic Map, EL Map, and PWLCM for 100 randomly selected values of the Control Parameter (a) with seed 0.0000000001 and (b) with Seed = 0.9999999999.

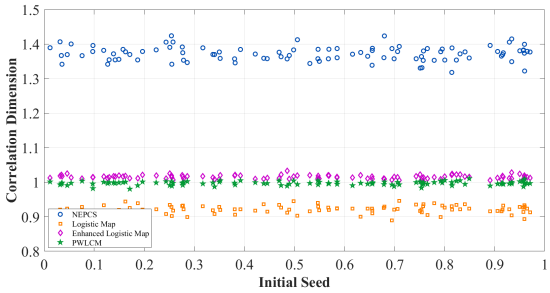



**FIGURE 8.** $D_2$ results for a fixed Control Parameter of NEPCS ($\lambda = 29.99$), Logistic Map ($\mu = 4$), EL Map ($\mu = 3.99$), and PWLCM ($\gamma = 0.49$) and 100 randomly selected values of Initial Seed.

be chaotic, the entropy value should be high, while a lower entropy value indicates less randomness [41].

To evaluate the chaotic behavior of the NEPCS in comparison with the Logistic map, Enhanced Logistic map, and PWLCM, the Sample Entropy is computed under varying conditions. The experiments consider the minimum value of the initial seed $x_0 = 0.0000000001$ along-with 100 randomly chosen values of the control parameters for all systems within specified ranges and results are given in Figures 9(a). Figure 9(b) displayed the results with same randomly selected control parameters but using the seed value $x_0 = 0.9999999999$. The sample entropy of the proposed system is far better than Logistic, EL map, and PWLCM.

Figure 10 presents the Sample Entropy results for fixed control parameters and 100 random values of the initial seed in the range of [0 - 1]. The maximum sample entropy value for NEPCS is 2.2127 and the minimum is 2.1533. The EL map has better values than Logistic and PWLCM with a minimum value of 1.9843 and a maximum value of sample entropy score of 2.0281, but all have lower scores than NEPCS.

The Sample Entropy score is computed for a fixed initial seed and control parameter, as detailed in Table 2. The findings indicate that the proposed system exhibits greater complexity and unpredictability than the existing chaotic maps.

#### 6) NIST SP800-22

The SP800-22 is a statistical test suite released by the National Institute of Standards and Technology (NIST) for

**TABLE 2.** Sample Entropy results for NEPCS ($\lambda = 29.99$), Logistic Map ($\mu = 4$), EL Map ($\mu = 3.99$), and PWLCM ($\gamma = 0.49$), initial Seed = 0.0000000001.

| PRNG | Sample Entropy |
|---|---|
| NEPCS | 2.2137 |
| Logistic Map | 0.6349 |
| Enhanced Logistic Map | 2.0043 |
| PWLCM | 0.4879 |

testing the empirical randomness of dynamical systems and encryption algorithms. This suite comprises 15 tests, each with a $p - value$ between 0 and 1. If the $p - value$ falls below a certain threshold, the system is considered stable; conversely, if it exceeds the threshold, it is deemed unstable and unpredictable [49].

In this study, the SP800-22 test suite is applied to 1000 distinct sequences, each of length $10^8$ generated using the NEPCS, and the average results are compared with the baseline chaotic maps as depicted in Table 3. Based on these results, it can be concluded that the proposed system demonstrates randomness, as it successfully passes all the tests with significant margins.

#### 7) TESTU01 ANALYSIS

The TestU01 was developed by L'Ecuyer and Simard in 2007 [50] and coded in C language. This test suite is designed for the empirical randomness testing of Random Number Generators (RNGs), and provides a comprehensive set of tests to evaluate the quality of RNGs. To assess the randomness of our proposed NEPCS, we utilized three test batteries of TestU01: Rabbit, Alphabit, and Block Alphabit. The two random sequences of length $2^{20}$ and $2^{30}$ have been produced via NEPCS using the same seeds and control parameters discussed in III-A, Logistic map, EL map, and PWLCM. The TestU01 results for these sequences, presented in Table 4, demonstrate that the proposed map is well-suited for pseudo-random number generation.

### B. SUMMARY

Based on the above analysis and results given in Table 5, it can be concluded the Normalized Piecewise Exponential

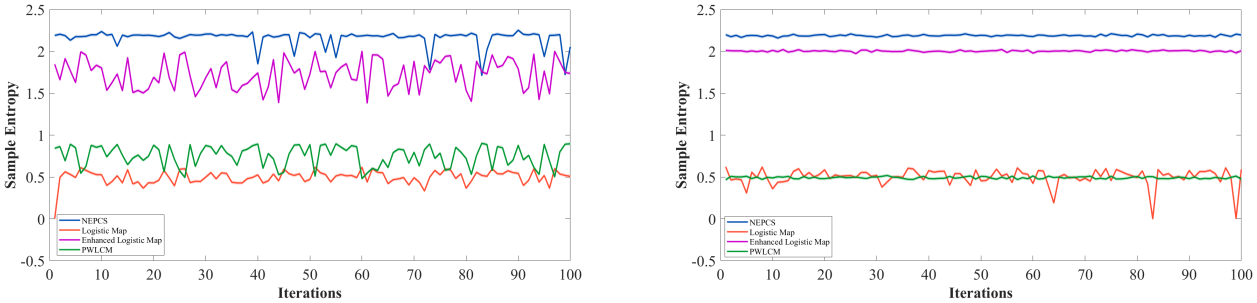



**FIGURE 9.** Sample Entropy results for Logistic Map, EL Map, and PWLCM for 100 randomly selected values of the Control Parameter (a) with seed 0.0000000001 and (b) with Seed = 0.9999999999.

**TABLE 3.** NIST SP800-22 results for Logistic Map, EL Map, Logistic Map, PWLCM, and NEPCS.

| S. No | Test Name | Logistic Map | EL Map | PWLCM | NEPCS |
|---|---|---|---|---|---|
| 1 | Frequency | 0.1238 | 0.1232 | 0.9940 | 0.4946 |
| 2 | Block Frequency | 0.2478 | 0.4521 | 0.4560 | 0.4901 |
| 3 | Cumulative Sums | 0.1316 | 0.8454 | 0.8560 | 0.5101 |
| 4 | Runs | 0.1029 | 0.1352 | 0.9240 | 0.4913 |
| 5 | Longest Run | 0.2468 | 0.5421 | 0.7200 | 0.4865 |
| 6 | Rank | 0.4094 | 0.3542 | 0.6160 | 0.4623 |
| 7 | FFT | 0.3463 | 0.4115 | 0.0400 | 0.4623 |
| 8 | Non-Overlapping Template | 0.4597 | 0.2154 | 0.4820 | 0.5072 |
| 9 | Overlapping Template | 0.4399 | 0.3214 | 0.9640 | 0.4980 |
| 10 | Universal | 0.0096 | 0.3214 | 0.8680 | 0.3345 |
| 11 | Approximate Entropy | 0.0109 | 0.7441 | 0.8680 | 0.2697 |
| 12 | Random excursions | 0.4925 | 0.3125 | 0.2660 | 0.9241 |
| 13 | Random excursions variant | 0.0020 | 0.3224 | 0.3080 | 0.8677 |
| 14 | Serial | 0.0020 | 0.1654 | 0.2690 | 0.4576 |
| 15 | Linear Complexity | 0.0000 | 0.8871 | 0.0460 | 0.4947 |

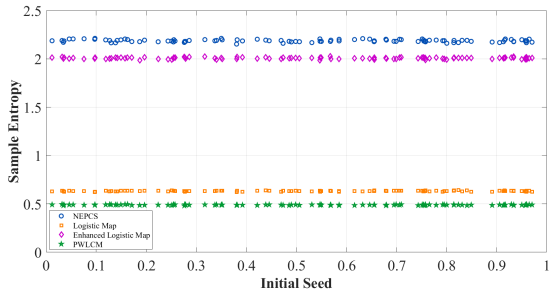



**FIGURE 10.** Sample Entropy results for a fixed Control Parameter of NEPCS ($\lambda = 29.98$), Logistic Map ($\mu = 4$), EL Map ($\mu = 3.99$), and PWLCM ($\gamma = 0.49$) and 100 randomly selected values of initial seed.

Chaotic System (NEPCS), exhibits higher $D_2$ and sample entropy scores, better uniform distribution, high sensitivity to initial conditions and enhanced randomness across a large key space for the control parameter ($-30 < \lambda < 30$) excluding [$-0.5$ to $+0.5$], with a precision of $10^{-12}$. These attributes make it particularly advantageous for practical applications. Furthermore, the system demonstrates superior pseudo-random characteristics compared to some of the existing 1D systems Logistic, EL, and PWLCM.

## IV. DNA COMPLEMENTARY RULES AND ALGEBRAIC OPERATIONS

Every sequence of Deoxyribose Nucleic Acid (DNA) comprises four nucleic acid bases: A, C, G, and T. The letter A illustrates Adenine, C illustrates Cytosine, G indicates Guanine, and T is for Thymine. There are $4! = 24$ possible ways of pairing four nucleotides, but only eight follow Watson-Crick complementary rules that are depicted in Table 6. The mathematical operations like Exclusive-OR, addition, and subtraction can be performed on the DNA Bases described in Tables 7 and 8.

## V. PROPOSED IMAGE ENCRYPTION SCHEME

A good cipher must be steeled toward various cryptographic attacks, and the algorithms should have a sound mathematical foundation. This has prompted the development of a new system that employs a permutation-substitution framework and DNA encoding and decoding to add complexity. This encryption system begins with a 24-bit RGB image and initial seeds $b_0, b_1, b_2, b_3, b_4, b_5, b_6$ for the NEPCS to generate pseudo-random numbers. Two more control parameters; $\lambda$ and prime number $p$, are also required from the user as keyboard inputs. The proposed encryption system uniquely blends keyless and keyed permutations before and after the substitution phase. The seeds and plain image are used intervolving under the SHA-256 function, and the final hash value is used to extract the seeds to hinder the chosen plaintext attacks.

### A. GENERATION OF KEYS

The first step in designing the image encryption algorithm involves processing the keyboard input so that a minute change in the input image automatically alters the

**TABLE 4.** TestU01 results for logistic Map, EL Map, PWLCM, and NEPCS.

| Battery | Test Parameters | Number of Tests | Sequence Size | Logistic | EL Map | PWLCM | NEPCS |
|---|---|---|---|---|---|---|---|
| Rabbit | Standard | 38 | $2^{20}$ | Pass* | Pass* | Pass* | Pass |
| | | | $2^{30}$ | Pass* | Pass* | Pass* | Pass* |
| Alphabit | Standard | 38 | $2^{20}$ | Pass* | Pass* | Pass* | Pass |
| | | | $2^{30}$ | Pass* | Pass* | Pass* | Pass* |
| Block Alphabit | Standard | 38 | $2^{20}$ | Pass* | Pass* | Pass* | Pass |
| | | | $2^{30}$ | Pass* | Pass* | Pass* | Pass* |

* Partially Qualified

**TABLE 5.** Comparative summary of Logistic, EL Map, PWLCM and NEPCS.

| Chaotic Map | Logistic Map | EL Map | PWLCM | NEPCS |
|---|---|---|---|---|
| Chaotic Range | [3.57 - 4] | [0 - 4] | [0 - 1] | [-30 - +30] |
| Lyapunov Exponent | 0.6937 | 3.6474 | 0.8543 | $> 7.88$ |
| Sample Entropy | 0.6263 | 1.9972 | 0.4724 | 2.2137 |
| Correlation Dimension | 0.9346 | 1.0185 | 0.9892 | 1.3735 |
| NIST Test Suite | Pass* | Pass | Pass | Pass |
| TestU01 | Pass* | Pass* | Pass* | Pass |
| Histogram Analysis | Non-uniform | Non-uniform | Non-uniform | Uniform |
| Complexity | $O(n)$ | $O(n)$ | $O(n)$ | $O(e^n)$ |
| Key Space | $2^{96}$ | $2^{102}$ | $2^{100}$ | $2^{187}$ |

**TABLE 6.** Complementary DNA rules.

| **Rule** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** |
|---|---|---|---|---|---|---|---|---|
| **00** | A | A | C | C | G | G | T | T |
| **01** | C | G | A | T | A | T | C | G |
| **10** | G | C | T | A | T | A | G | C |
| **11** | T | T | G | G | C | C | A | A |

**TABLE 7.** Exclusive-OR operations for DNA bases.

| $\oplus$ | **A** | **G** | **C** | **T** |
|---|---|---|---|---|
| **A** | A | G | C | T |
| **G** | G | A | T | C |
| **C** | C | T | A | G |
| **T** | T | C | G | A |

**TABLE 8.** Addition and subtraction operations for DNA bases.

| **+** | **A** | **G** | **C** | **T** | **-** | **A** | **G** | **C** | **T** |
|---|---|---|---|---|---|---|---|---|---|
| **A** | A | G | C | T | **A** | A | G | T | C |
| **G** | G | A | T | C | **G** | G | A | C | T |
| **C** | C | T | G | A | **C** | C | T | A | G |
| **T** | T | C | A | G | **T** | T | C | G | A |

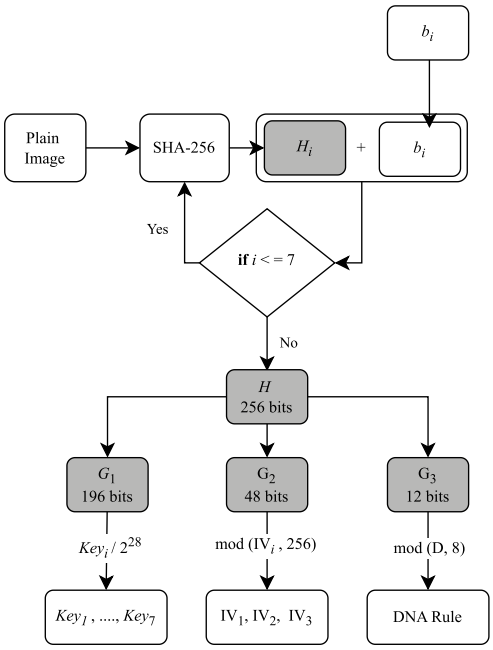



**FIGURE 11.** Key generation process.

corresponding key values to resist chosen-plaintext attacks. The theme of the proposed image cipher requires seven vectors of pseudo-random numbers, which in turn requires seven different initial seeds input by the user called $b_0, b_1, b_2, b_3, b_4, b_5$ and $b_6$. These keyboard inputs are processed along-with the 256-bit hash $H_0$ of the plain image and it is concatenated with $b_0$ to fed into the SHA-256 to compute another message digest called $H_1$. The similar mechanism of concatenation and creating $H_i$ continues six more times and finally receives $H_7$. The message digest $H_7$ is split into three groups: $G_1$ of size 196-bit, $G_2$ of size 48-bit, and $G_3$ of size 12-bit. The entire process is summarized in Figure 11.

The first group $G_1$ is divided into seven subgroups called $key_i$ where $i = 1, 2, \ldots, 7$; each of 28 bits. In Equation (11), the $key_i$ are cast to floating point precision value ranging from 0 to 1 with precision of $10^{-15}$. Similarly, $G_2$ is split into three parts, each of 16-bit in length called $IV_1, IV_2, IV_3$, and processed to get values in the range of [0-255]. The

processing is represented mathematically in Equation (12). This method helps to ensure that the secret key is a complex function of the plain image to enhance security, hence covering the problem of chosen plaintext attacks.

$$key_i = \frac{bin2dec\,(key_i)}{2^{28}} \tag{11}$$

$$IV_i = bin2dec\,(IV_i) \bmod 256 \tag{12}$$

The last group $G_3$ is also casted into a value in the range of [0-7] for the selection DNA rule for decoding shown in Equation (13).

$$D = bin2dec\,(D) \bmod 8 \tag{13}$$

### B. INITIAL PERMUTATION

The heart of the shuffling process is the operation of interchanging two pixels to move them away from their original locations and weaken the correlation in all paths (horizontal, vertical and diagonal) of the image. The permutation method employed in the proposed algorithm dependent on pseudo-random numbers generated by NEPCS; higher randomness leads to a more weak correlations. The 24-bit RGB image denoted as $I$ of size $MN$ is the input to the encryption scheme.

#### 1) STEP 1

The RGB image $I$ is split into channels $I_r$, $I_g$ and $I_b$. A vector $\mathcal{A}$ of size $1 \times 3MN$ is created to store all the pixels of channels as shown in Equation (14).

$$\begin{cases} \mathcal{A}_{(1 \longrightarrow MN)} = I_r \\ \mathcal{A}_{(MN+1 \longrightarrow 2MN)} = I_g \\ \mathcal{A}_{(2MN+1 \longrightarrow 3MN)} = I_b \end{cases} \tag{14}$$

#### 2) STEP 2

A pseudo random sequence $\mathcal{R}_1$ of size $1 \times (3MN + t)$ is generated by iterating Equation (4) with initial condition $key_1$. The $t$ number of elements are eliminated to prevent the transient effects. The $\mathcal{R}_1$ is sorted as depicted in Equation (15).

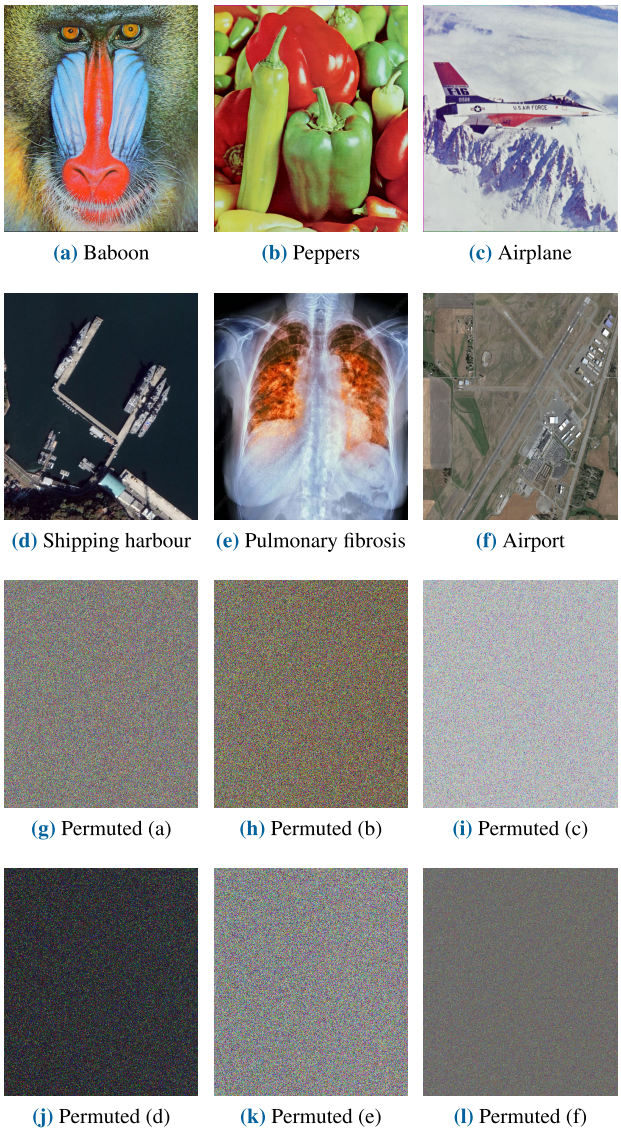
$$\left[\mathcal{R}_1^{idx},\, \mathcal{R}_1^{acs}\right] = Sort\,(\mathcal{R}_1) \tag{15}$$

The $\mathcal{R}_1^{asc}$ is sorted vector in ascending order and $\mathcal{R}_1^{idx}$ hold indexes of sorted values of $\mathcal{R}_1^{asc}$ sequence.

#### 3) STEP 3

The index vector $\mathcal{R}_1^{idx}$ is used to shuffle the position of the pixels to complete the permutation. The inputs are $\mathcal{A}$, sorted index vector $\mathcal{R}_1^{idx}$ and output is represented mathematically as $\hat{A}$ represented mathematically in (16). Figure 12 shows the result of permutation for different images and each image has no visual information; hence weakens the correlation in all three directions.

$$\hat{A} = Permute(\mathcal{A}, \mathcal{R}_1^{idx}) \tag{16}$$



(a) Baboon (b) Peppers (c) Airplane
(d) Shipping harbour (e) Pulmonary fibrosis (f) Airport
(g) Permuted (a) (h) Permuted (b) (i) Permuted (c)
(j) Permuted (d) (k) Permuted (e) (l) Permuted (f)

**FIGURE 12.** Permutation analysis: Plain Images: (a) - (f); Permutated Images: (g) - (l);.

### C. DNA ENCODING

For DNA encoding of pixels, the permuted vector $\hat{A}$ is split into three vectors called $\hat{A}_r$, $\hat{A}_g$ and $\hat{A}_b$ where each having the size of $MN \times 1$. So, before proceeding to substitution, DNA encoding of the pixels of all three vectors $\hat{A}_r$, $\hat{A}_g$ and $\hat{A}_b$ are required.

#### 1) PSEUDO-RANDOM NUMBER GENERATION

The DNA encoding in an arbitrarily way requires three vectors of random numbers $\mathcal{R}_2$, $\mathcal{R}_3$ and $\mathcal{R}_4$. The Equation (4) is revisited to generate sequence of $MN + t \times 1$ using the initial seeds $key_2$, $key_3$ and $key_4$. The operation of modulus 4 is applied on each sequence to adjust the sequence scale between 0 to 3 as illustrated in Equation (17) to select one of four DNA rules to encode for a pixel.

$$\bar{\mathcal{R}}_2 = round(\mathcal{R}_2 \times 10^{16}) \bmod 4$$

$$\bar{\mathcal{R}}_3 = round(\mathcal{R}_3 \times 10^{16}) \bmod 4$$
$$\bar{\mathcal{R}}_4 = round(\mathcal{R}_4 \times 10^{16}) \bmod 4 \quad (17)$$

#### 2) DNA ENCODING OF PIXELS

For DNA encoding, the permuted vectors are converted into 8-bit binary which is a necessary step in translating bits into DNA bases. The mathematical expression is shown in Equation (18). The resultant binary matrix has $MN \times 8$.

$$\dot{A}_{r_{(MN\times 8)}} = \hat{A}_r$$
$$\dot{A}_{b_{(MN\times 8)}} = \hat{A}_b$$
$$\dot{A}_{b_{(MN\times 8)}} = \hat{A}_g \quad (18)$$

The binary of each color pixel is transformed into DNA bases by selecting one of the four complementary DNA rules shown in Table 9. This selection is dependent on the pseudo-random vectors $\bar{\mathcal{R}}_2$, $\bar{\mathcal{R}}_3$ and $\bar{\mathcal{R}}_4$ for $\dot{A}_r$, $\dot{A}_g$ and $\dot{A}_b$. The DNA rule is represented by $Rule_j$ where $j$ has numeric values of 0, 1, 2, and 3. The Algorithm # 1 represents the complete process of conversion of pixels into binary, and then into DNA bases for $\dot{A}_r$ and the output is $E_r$ with a size of $M \times 4N$. The Algorithm # 1 is also deployed on $\dot{A}_g$, $\dot{A}_b$ using random sequences $\bar{\mathcal{R}}_3$, $\bar{\mathcal{R}}_4$ to obtain $E_g$ and $E_b$ respectively.

### D. SUBSTITUTION

For designing a secure cipher, the permutation only phase is not enough to achieve the encryption as the statistical information remains the same, and an attacker can use this information to cryptanalyses it easily. The substitution plays a pivotal role in designing secure ciphers, and the Cipher Block Chaining (CBC) mode is sandwiched in between two keyless transpositions ZigZag and Snake. Hence, substitution consisted of three steps, ZigZag transaction, addition and exclusive-or operations on DNA bases and Snake keyless transposition technique.

#### 1) ZIGZAG TRANSPOSITION

Before implementing the CBC encryption mode on DNA bases, a keyless transposition method known as ZigZag is applied to the DNA bases. This keyless transposition works as an inter-pixel substitution making the proposed cipher more complex and obtaining the final output as 1D vector. The traversing pattern of the zigzag scans the atomic data unit from the top left corner to the bottom right corner, following the direction of diagonal order. This zigzag pattern is diagrammatically displayed in Figure 13 and applied on the encoded color channel $E_r$, $E_g$ and $E_b$. All elements are stored in three separate 1D vectors $\mathcal{Z}_r$, $\mathcal{Z}_g$, and $\mathcal{Z}_b$, respectively.

#### 2) CBC ENCRYPTION

So far, Zigzag transposition has been introduced, which exchanges the position of DNA bases and weakens the statistical information at the 2-bit level. In the CBC substitution mode, the exclusive-or and addition operations are employed on DNA bases to employ the substitution phase of the proposed cipher. How the alternation of DNA addition and XOR operations work on the block of 4-DNA bases representing a pixel at each step are shown in Tables 7 and 8.

**TABLE 9.** Selected DNA rules for encoding.

| Rule | $Rule_0$ | $Rule_1$ | $Rule_2$ | $Rule_3$ |
|---|---|---|---|---|
| **00** | A | T | G | C |
| **01** | C | G | A | T |
| **10** | G | C | T | A |
| **11** | T | A | C | G |

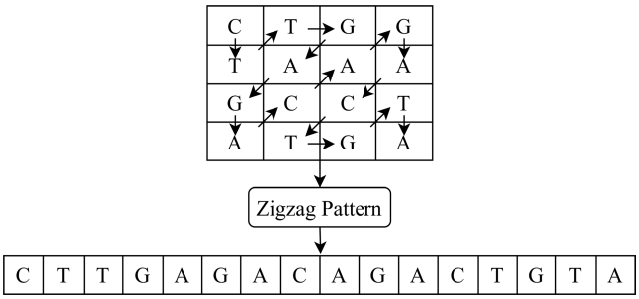



**FIGURE 13.** Zigzag pattern.

For CBC, three initial vectors $IV_1$, $IV_2$ and $IV_3$ are created in the range of [0-255] during the key generation process from $G_2$ and then $IV_i$ are transformed into DNA bases using any DNA rules depicted in Table 6. For the ease of implementation of substitution, the $\mathcal{Z}_r$, $\mathcal{Z}_g$ and $\mathcal{Z}_b$ are reshaped into 1D vector of size $1 \times 4MN$. The four DNA bases of $IV_1$ are used to substitute the first group of 4 DNA bases of vector $\mathcal{Z}_r$ under exclusive-or operation. The Algorithm # 2 is the pseudo-code of CBC mode of substitution in which both additions and exclusive-or($+, \oplus$) are used interchangeably as shown in Figure 14. The $1^{st}$ group of 4 DNA bases is substituted with $IV_1$ using exclusive-or operation and the $2^{nd}$ group of 4 DNA bases is added with encrypted DNA bases of $1^{st}$ group, $3^{rd}$ is exclusive-ored with $2^{nd}$ group of encrypted DNA bases and so on.

The Algorithm # 2 is also applied on $\mathcal{Z}_g$ and $\mathcal{Z}_b$ to obtain $\hat{Z}_g$ and $\hat{Z}_b$.

#### 3) SNAKE TRANSPOSITION

The snake transposition is another keyless transposition technique which can be applied on the matrix of DNA bases. So to make it applicable, the output of the CBC substitution is transformed into a DNA matrix of size $M \times 4N$. This technique fills the vector serpentine manner, inducing another layer of complexity to the cipher design. The graphical representation of the snake pattern is depicted in Figure 15 and applied on $\hat{Z}_r$, $\hat{Z}_g$ and $\hat{Z}_b$ and the output of the snake pattern is denoted by $\mathcal{S}_r$, $\mathcal{S}_g$ and $\mathcal{S}_b$ and finally substitution is completed.

### E. FINAL PERMUTATION

The final permutation occurs on the DNA bases, where each substituted 1D vector of DNA bases is shuffled using the sorted index of the random numbers generated from NEPCS.

**Algorithm 1** DNA Encoding Algorithm

**procedure** DNAENCODE($\dot{A}_r$, $Rule_j$, $\bar{\mathcal{R}}_2$, $M$, $N$)
  $l \leftarrow 1$ ▷ Initialize $l$ to 1 for sequential selection of pixel
  **for** $i \leftarrow 1$ to $M$ **do**
    $k \leftarrow 1$ ▷ Set to $k = 1$, initial index to save DNA bases in a new row each time
    **for** $kk \leftarrow 1$ to $N$ **do** ▷ Introduced loop to build 2D matrix
      $Rule \leftarrow Rule_j(\bar{\mathcal{R}}_2(l))$ ▷ Use $\mathcal{R}_2$ for selection of DNA rule $j$ is 0, 1, 2 or 3
      $E_r(i, k : k+3) \leftarrow$ **bin2dna**$[\dot{A}_r(l, :), Rule]$ ▷ Convert binary to 4DNA bases using randomly selected Rule
      $k \leftarrow k + 4$ ▷ Increment of 4 in $k$ for storing 4 DNA bases in same row of Matrix $E_r$
      $l \leftarrow l + 1$ ▷ Increment in $l$ to continue $MN$ times for sequential selection of pixel's binary and $\mathcal{R}_2$
    **end for**
  **end for**
**end procedure**
$Output = E_r$ ▷ 2D matrix of DNA bases having size of $M \times 4N$

**Algorithm 2** CBC Substitution

**procedure** CBC SUBSTITUTION($IV_1$, $\mathcal{Z}_r$, $M$, $N$)
  $\hat{Z}_r(1:4) = \mathcal{Z}_r(1:4) \oplus IV_1$
  $k \leftarrow 5$
  **for** $i \leftarrow 2$ to $MN/2$ **do**
    $\hat{Z}_r(k : k+3) = \hat{Z}_r(k-4 : k-1) + \mathcal{Z}_r(k, k+3)$
    $\hat{Z}_r(k+4 : k+7) = \hat{Z}_r(k : k+3) \oplus \mathcal{Z}_r(k+4, k+7)$
    $k \leftarrow k + 8$
  **end for**
**end procedure**

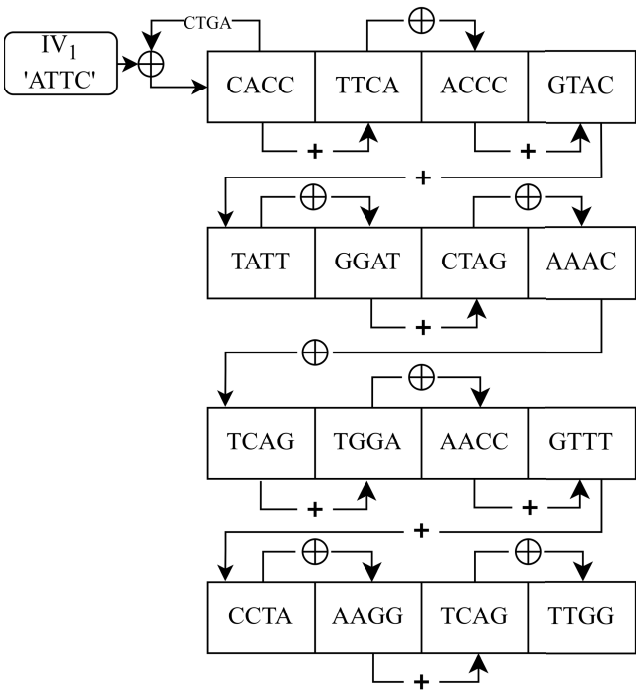



**FIGURE 14.** CBC mode for DNA substitution.

### 1) STEP 1

For this permutation, three pseudo random sequences referred as $\mathcal{R}_5$, $\mathcal{R}_6$ and $\mathcal{R}_7$ are generated by iterating Equation (4) with initial conditions $key_5$, $key_6$ and $key_7$. The size of each vector

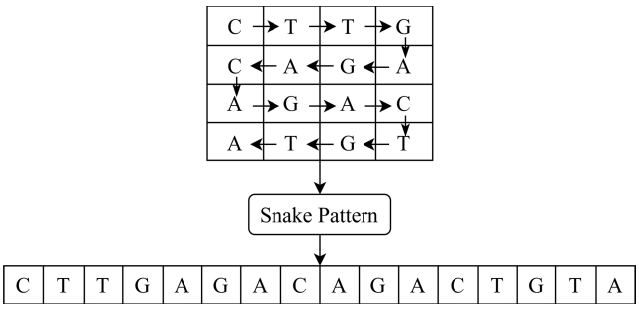



**FIGURE 15.** Snake left hand pattern.

is $1 \times 4MN + t$ and discard the first $t$ number of elements to eliminate the transient effect. The pseudo-random vectors $\mathcal{R}_5$, $\mathcal{R}_6$ and $\mathcal{R}_7$ are sorted and record their index as shown in Equation (19).

$$\begin{aligned}\left[\mathcal{R}_5^{idx}, \mathcal{R}_5^{acs}\right] &= Sort\,(\mathcal{R}_5)\\ \left[\mathcal{R}_6^{idx}, \mathcal{R}_6^{acs}\right] &= Sort\,(\mathcal{R}_6)\\ \left[\mathcal{R}_7^{idx}, \mathcal{R}_7^{acs}\right] &= Sort\,(\mathcal{R}_7)\end{aligned} \tag{19}$$

The $\mathcal{R}^{asc}$ has sorted values of $\mathcal{R}$ sequence in ascending order and $\mathcal{R}^{idx}$ hold indexes of sorted values of $\mathcal{R}^{asc}$ sequence.

### 2) STEP 2

The index vectors $\mathcal{R}^{idx}$ are used for disturbances between the position of DNA bases to complete the shuffling process. The $\hat{S}_r$, $\hat{S}_g$ and $\hat{S}_b$, sorted index vectors $\mathcal{R}_5^{idx}$, $\mathcal{R}_6^{idx}$ and $\mathcal{R}_7^{idx}$ are the inputs to the final permutation process shown in (20) while the outputs are $\hat{\mathcal{E}}_r$, $\hat{\mathcal{E}}_g$ and $\hat{\mathcal{E}}_b$ as display in Equation (20).

$$\begin{aligned}\hat{\mathcal{E}}_r &= Permute(\hat{S}_r, \mathcal{R}_5^{idx})\\ \hat{\mathcal{E}}_g &= Permute(\hat{S}_g, \mathcal{R}_6^{idx})\\ \hat{\mathcal{E}}_b &= Permute(\hat{S}_b, \mathcal{R}_7^{idx})\end{aligned} \tag{20}$$

## F. DNA DECODING

The vectors $\hat{\mathcal{E}}_r$, $\hat{\mathcal{E}}_g$ and $\hat{\mathcal{E}}_b$ are decoded back into decimal number by adopting DNA rule $D$ computed during key

generation process. The $\tilde{E_r}$, $\tilde{E_g}$ and $\tilde{E_b}$ represents resultant channels of size $MN$, and illustrated in Equation (21).

$$\tilde{E_r} = Decode(E_r, D)$$
$$\tilde{E_g} = Decode(E_g, D)$$
$$\tilde{E_b} = Decode(E_b, D) \quad (21)$$
$$E = cat(\tilde{E_r} + \tilde{E_g} + \tilde{E_b}) \quad (22)$$

All channels are articulate to form encrypted image $E$ as the final output of encryption as illustrated in Equation (22). The Figure 16 demonstrates the complete image encryption process.

### G. DECRYPTION PROCESS

In symmetric encryption, the decryption process from the last step uses the same secret keys to return the original data. The following simple steps involved in the decryption

- The RGB cipher $E$ split into channels $E_r$, $E_g$ and $E_b$ opposite to the Equation (22).
- The encoding of the channels using DNA rule $D$ chosen from Table 6 is utilized as displayed via Equation (21). After this, the encoded channels are converted into 1D vectors.
- The inverse of permutation of DNA bases with the aid of sorted indexes of random number $\mathcal{R}_5$, $\mathcal{R}_6$ and $\mathcal{R}_7$ are employed as exhibited in Equation (19) and Equation (20).
- Before CBC substitution, inverse snake pattern is applied on each vector.
- The inverse CBC substitution mode is applied under XOR and subtraction operation on the permuted DNA vectors as represented in Algorithm # 3 and graphically represented in Figure 14. The last group of DNA bases is subtracted from the $2^{nd}$ last group of DNA bases and will continue till the $2^{nd}$ group of DNA bases. The first group of 4DNA bases will be decrypted using corresponding $IV_i$.
- Final step of substitution process is the to apply the inverse of the ZigZag pattern.
- The sorted indexes of random number $\mathcal{R}_2$, $\mathcal{R}_3$ and $\mathcal{R}_4$ are utilized to select DNA rules randomly to decode back to binary and then to decimal format. The decoded vectors are combined to form a single vector.
- The initial permutation is performed using the sorted index vector of random number $\mathcal{R}_1$, and the permuted vector is transformed into deciphered RGB channels. All channels are combined to form a decrypted image.

## VI. SIMULATION AND RESULTS

The trials to investigate the encryption algorithm are performed on a personal laptop with Core i5 11300H CPU with a frequency of 2.50GHz and 16 GB of RAM with MATLAB R2023b. The image database created at Signal and Image Processing Institute based at the University of Southern California (USC-SIPI) is used. USC-SIPI images Baboon, Peppers, and Airplane of size 512 × 512 were employed in the experiments. Three images, Shipping harbour [51], Pulmonary fibrosis at https://www.sciencephoto.com/media/880215/view and Airport [52] are randomly chosen to analyze the strength of the proposed encryption method.

For the experiments, the initial seeds for NEPCS are assumed as $b_0 = 0.381294613786$, $b_1 = 0.749456435562$, $b_2 = 0.567618257948$, $b_3 = 0.925832879324$, $b_4 = 0.613786435562$, $b_5 = 0.257948879324$ and $b_6 = 0.437912567618$ which are concatenated with the digest of plain image iteratively to produce $key_i$, $IV_j$ and $D$. The control parameters $\lambda = 3.123$, prime number $p = 367$. The set of all the initial seeds, $p$, and $\lambda$ are collectively called $\delta_0$. When modifying any one of the initial seed $b_i$, $\lambda$ or $p$ will create a new key set $\delta_1$. This allows for a total of nine key sets $\delta_1$, $\delta_2$, $\delta_3$, $\delta_4$, $\delta_5$, $\delta_6$, $\delta_7$. $\delta_8$, and $\delta_9$. In the substitution phase, three initial vectors $IV_1$, $IV_2$ and $IV_3$ are encoded by selecting one of rule $Rule_j$ from Table 6 at this step of the cipher.

Figure 17 illustrates the encryption results of the proposed cipher for different plain images which are shown in Figure 12(a) - 12(f) while Figure 17(g) - 17(l) is the result of the decryption process. Nevertheless, an encryption algorithm's security must be proved through statistical scrutiny. Different metrics are employed to evaluate statistical, differential, key space and key sensitivity analysis, noise robustness and time efficiency analysis to investigate the quality of the encryption system.

### A. STATISTICAL ANALYSIS

#### 1) NIST SP800-22 TESTS

The description of the NIST SP800-22 tests has been discussed in the section III-A6. The encrypted images shown in Figure 17(a) to 17(f) are converted into binary files and tested for the NIST test suit. The results of these images are given in Table 10. All the images successfully qualified for the NIST test with a probability score $p >$ greater than 0.05.

#### 2) 3D HISTOGRAM ANALYSIS

A histogram depicts the frequency of each gray level of an image and plays a vital role in evaluating an image encryption algorithm. The cryptanalyst can employ the frequency information from the histograms of plain and encrypted images to break the cipher without knowing the secret key. The effectiveness of the image encryption algorithm depends on its ability to eliminate any visible patterns or statistical redundancies in the original image. A uniformly distributed histogram proves that image encryption conceals the statistical information. Figure 18(a) illustrates the 3D histograms of the plain baboon image while Figure 18(b) and Figure 18(c) represent the histograms of the ALL black and All white images respectively. The Figure 18(d)-(e) are the histograms of encrypted images of Baboon, All black and All white. The encrypted images have a fairly uniform

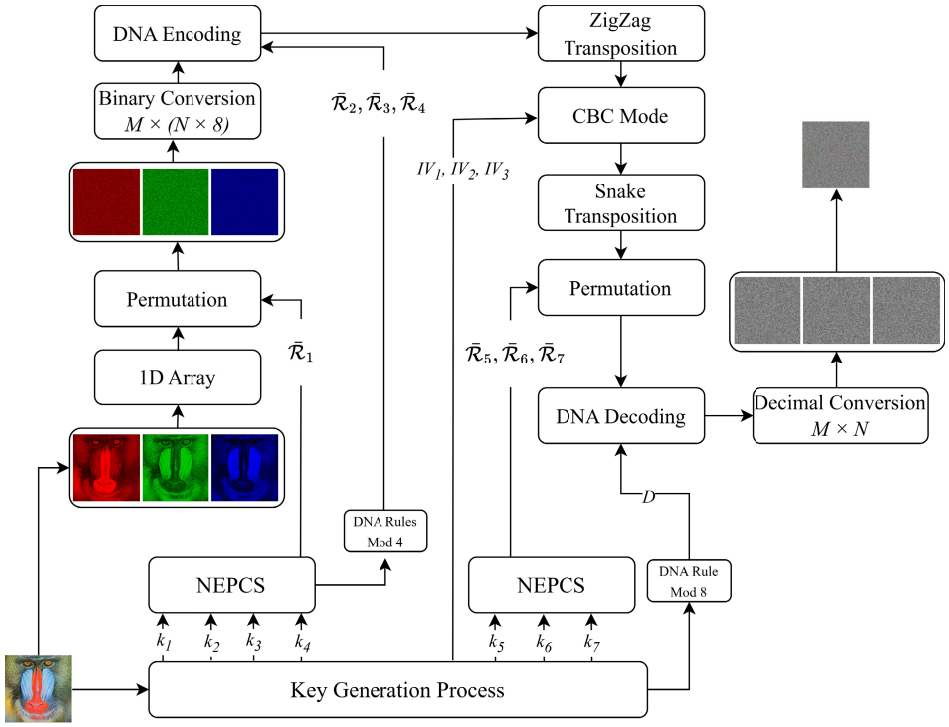



**FIGURE 16.** Flow diagram of proposed encryption scheme.

**Algorithm 3** CBC Inverse Substitution

**procedure** CBC INV SUBSTITUTION($IV_1, \hat{S}_r, M, N$)
  $k \leftarrow 4MN$ ▷ Begin the process from last 4 DNA bases
  **for** $i \leftarrow 2$ to $MN/2$ **do**
    $\mathcal{S}_r(k-3:k) = \hat{S}_r(k-3:k) \oplus \hat{S}_r(k-7:k-4)$ ▷ Last 4 DNA bases decrypted
    $\mathcal{S}_r(k-11:k-8) = \hat{S}_r(k-7:k-3) - \hat{S}_r(k-11:k-8)$ ▷ Decryption of 2nd Last group of 4DNA bases
    $k \leftarrow k-8$
  **end for**
  $\mathcal{S}_r(1:4) = \hat{S}_r(1:4) \oplus IV_1$ ▷ Decryption of 1st group of 4DNA bases
**end procedure**

distribution, proving the proposed system's strength. The size of a sphere represents the frequency of a gray level in red, green, and blue channels.

### 3) CORRELATION COEFFICIENT

A correlation coefficient test is capable of confirming the robustness of the image encryption algorithm in that it successfully removes the statistical properties observed in the plain image. The Equation (23) is used to compute the correlation coefficient as follows:

$$\rho_{xy} = \frac{|Cov(x,y)|}{\sqrt{D(x) \times D(y)}}$$

$$Cov(x,y) = \frac{1}{N}\sum_{i=1}^{N}(x_i - E(x))\,(y_i - E(y))$$

$$E(x) = \frac{1}{N}\sum_{i=1}^{N} x_i$$

$$D(x) = \frac{1}{N}\sum_{i=1}^{N}(x_i - E(x))^2 \tag{23}$$

where $N$ stands for the number of pairs of randomly chosen pixel points, $x$ and $y$ define gray values of adjoining pixels, $D(x)$ denotes sample variance, $E(x)$ refers to sample expectation, and $p_{xy}$ defines the coefficient. The correlation coefficient computed via Equation (23) lies within the range of $+1$ to $-1$. The higher the coefficient score, the stronger the correlation among adjacent pixels, and vice versa. Generally, the correlation coefficient of the plain image will be nearly 1 due to the high correlation among neighboring pixels. In contrast, an encrypted image should have a coefficient score close to 0 or less.

Figure 19 is presents the scatter plots of 3000 pixels chosen randomly in all three directions; horizontal, vertical and diagonal. A dot in a plane of the scatter plot depicts the gray values of a pair of adjacent pixels selected randomly.

**TABLE 10.** Analysis of NIST test for encrypted images.

| S. No. | Test Name | Baboon | Peppers | Airplane | Shipping harbour | Pulmonary fibrosis | Airport |
|---|---|---|---|---|---|---|---|
| 1 | Frequency | 0.1223 | 0.7399 | 0.06301 | 0.5342 | 0.9114 | 0.7399 |
| 2 | Block Frequency | 0.2133 | 0.7399 | 0.1223 | 0.7399 | 0.7399 | 0.7399 |
| 3 | Cumulative Sums | 0.2133 | 0.9114 | 0.7399 | 0.9114 | 0.7399 | 0.5342 |
| 4 | Runs | 0.5342 | 0.5342 | 0.9114 | 0.9114 | 0.7399 | 0.7399 |
| 5 | Longest Run | 0.3505 | 0.3505 | 0.5342 | 0.5342 | 0.7399 | 0.3505 |
| 6 | Rank | 0.5342 | 0.7399 | 0.9114 | 0.7399 | 0.3505 | 0.7399 |
| 7 | FFT | 0.3505 | 0.5342 | 0.7399 | 0.3505 | 0.5345 | 0.7399 |
| 8 | Non-Overlapping Template | 0.9114 | 0.7399 | 0.3505 | 0.5342 | 0.9915 | 0.9114 |
| 9 | Overlapping Template | 0.9114 | 0.7399 | 0.9114 | 0.7399 | 0.5342 | 0.2133 |
| 10 | Approximate Entropy | 0.0888 | 0.0888 | 0.2133 | 0.2133 | 0.0669 | 0.9114 |
| 11 | Random excursions | 0.8317 | 0.8232 | 0.7541 | 0.9050 | 0.8832 | 0.6953 |
| 12 | Random excursions variant | 0.8656 | 0.8346 | 0.9148 | 0.9834 | 0.9620 | 0.9182 |
| 13 | Serial | 0.7399 | 0.7399 | 0.7399 | 0.9114 | 0.7399 | 0.9114 |
| 14 | Linear Complexity | 0.7399 | 0.2133 | 0.0846 | 0.7399 | 0.7399 | 0.7399 |

**TABLE 11.** Score of correlation coefficient for different images and comparison.

| Image | Channel | Plain Image | | | Encrypted Image | | |
|---|---|---|---|---|---|---|---|
| | | Horizontal | Vertical | Diagonal | Horizontal | Vertical | Diagonal |
| Baboon | Red | 0.9277 | 0.8675 | 0.8442 | -0.0145 | -0.0275 | -0.0066 |
| | Green | 0.8618 | 0.7749 | 0.7177 | -0.0133 | -0.0342 | -0.0565 |
| | Blue | 0.9060 | 0.8730 | 0.8460 | -0.0216 | -0.0028 | -0.0137 |
| Peppers | Red | 0.9602 | 0.9613 | 0.9506 | -0.0185 | -0.0221 | -0.0092 |
| | Green | 0.9817 | 0.9751 | 0.9736 | -0.0129 | -0.0199 | -0.0150 |
| | Blue | 0.9656 | 0.9621 | 0.9451 | -0.0210 | -0.0021 | -0.0042 |
| Airplane | Red | 0.9722 | 0.9549 | 0.9264 | -0.0006 | -0.0482 | -0.0030 |
| | Green | 0.9635 | 0.9725 | 0.9256 | -0.0093 | -0.0137 | -0.0124 |
| | Blue | 0.9641 | 0.9288 | 0.8895 | -0.0182 | -0.0030 | -0.0065 |
| Peppers [53] | Red | 0.9267 | 0.8757 | 0.8628 | 0.0016 | 0.0008 | -0.0032 |
| | Green | 0.8659 | 0.7770 | 0.7459 | -0.0001 | -0.0011 | 0.0057 |
| | Blue | 0.9099 | 0.8858 | 0.8467 | -0.0002 | -0.0001 | -0.0116 |
| Airplane [54] | Red | 0.8677 | 0.9211 | 0.8581 | 0.0041 | 0.0081 | 0.0027 |
| | Green | 0.7793 | 0.8595 | 0.8278 | 0.0018 | -0.0005 | 0.0285 |
| | Blue | 0.8770 | 0.9058 | 0.7278 | 0.0011 | 0.0034 | -0.0228 |
| Baboon [55] | Red | 0.8596 | 0.9241 | 0.8553 | -0.0039 | 0.0018 | 0.0012 |
| | Green | 0.7672 | 0.8645 | 0.7364 | -0.0020 | 0.0007 | -0.0019 |
| | Blue | 0.8856 | 0.9066 | 0.8389 | -0.0078 | -0.0032 | 0.0073 |
| Peppers [56] | Red | 0.9599 | 0.6944 | 0.9268 | 0.0762 | 0.0665 | 0.0671 |
| | Green | 0.9487 | 0.9827 | 0.9626 | 0.0875 | 0.0787 | -0.0751 |
| | Blue | 0.9556 | 0.9627 | 0.9237 | 0.0405 | 0.0598 | 0.0637 |

The Figure 19(a)-(c) is for plain image of baboon, and Figure 19(d)-(f) is for the encrypted image in all three directions.

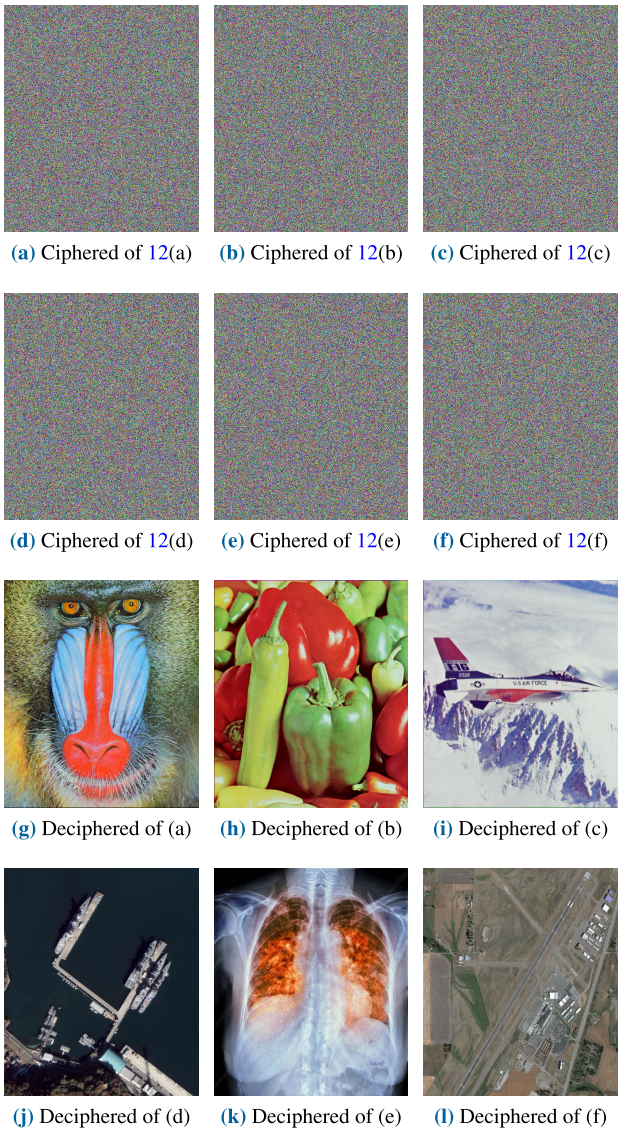
The correlation coefficient is computed for multiple images of plain and encrypted and demonstrated in Table 11. One can see that the coefficients of plain images are nearly 1, with a linear distribution center, while encrypted images reveal coefficients very close to [0, −1]. The evaluation demonstrates that the encryption algorithm within this study can successfully lower the correlation among adjacent pixels. We also made a comparison involving a range of different algorithms [53], [54], [55], [56].

#### 4) INFORMATION ENTROPY

In information theory, entropy refers to a measure of the magnitude of randomness in data. The Equation (24) can be used to calculate the degree of uncertainty (entropy) present in digital images [49].

$$H(s) = \sum_{i=0}^{L-1} p(si) \log_2 \frac{1}{p(si)} \tag{24}$$

Here, $L$ is the gray level and $p(si)$ signifies the probability of data occurrence of $L$ in the image. When calculating the entropy score of an image having a max gray level of 256, the theoretical entropy value must be close to 8. Lower entropy values imply that the encryption scheme has low randomness. The low entropy score could be a clear sign that the cipher may be prone to security breaches. The entropy values obtained for RGB channels for encrypted images are provided in Table 12. In addition, entropy results have been compared with several techniques from literature [9], [30], [53], [54], concentrating on colored images from encryption methods as presented in the Table 12 below.



(a) Ciphered of 12(a) (b) Ciphered of 12(b) (c) Ciphered of 12(c)

(d) Ciphered of 12(d) (e) Ciphered of 12(e) (f) Ciphered of 12(f)

(g) Deciphered of (a) (h) Deciphered of (b) (i) Deciphered of (c)

(j) Deciphered of (d) (k) Deciphered of (e) (l) Deciphered of (f)

**FIGURE 17.** Proposed encryption: Encrypted Outputs: (a) - (f); Decrypted Outputs: (g) - (l).

#### 5) LOCAL SHANNON ENTROPY

Local Shannon entropy (LSE) is a method to test the randomness in information at the local level, which is better organized and compatible with the cryptography of the images. Following is the mathematical description of the system to compute the LSE.

$$H_{k,T_B}(s) = \sum_{i=1}^{k} \frac{H(s_i)}{k} \tag{25}$$

In Equation (25), $S_i$ represents a randomly selected block with $k$ number of blocks such that $i \in 1, 2, \ldots, k$. In the above Equation (25), $T_B$ exhibits block size, and $H_k$ is averaging entropy of a $k$ randomly selected block of an image. The $k = 32$ and $T_B = 1936$ are used for the computation of LSE. The standard score of LSE for 8-bit gray image must be in the range between 7.9018 and 7.9034 [5], [57] while the standard value of LSE is 7.902469 under significance level $\alpha_{0.05}$. Table 13 displays the LSE score of encrypted images, and it can be witnessed that the proposed encryption scheme can obtain an acceptable LSE score and supersede the existing algorithm [26], [53], [55].

#### 6) PEARSON'S CHI-SQUARE TEST

The $\chi^2$ test can be employed to assess the histogram's uniformity, calculated as follows:

$$\chi^2 = \sum_{k=1}^{256} \frac{(v_k - o_k)^2}{256} \tag{26}$$

In Equation (26), notation $k$ refers to intensity level, while attributes $v$ and $o$ refer to observed and expected frequencies of gray value, respectively. The $\chi^2$ is employed to assess uniformity, and the results are presented in Table 14 for different images. The $\chi^2$ score must qualify the critical value $\chi^2_\alpha$, where $\alpha$ is known as the significance level and demonstrates the degree of freedom for $\chi^2$ distribution. For $\alpha = 0.05$, critical $\alpha$ value is computed as $\chi_{0.05} = 293.247$ [4]. According to the results, distribution of each ciphered image is quite close to uniform one at significance level as compared with Ref. work of [23], [29], [53], [55]. This then implies that histogram-based attacks are difficult to perform. The algorithm has fully explored statistical analysis and resilience against various attacks.

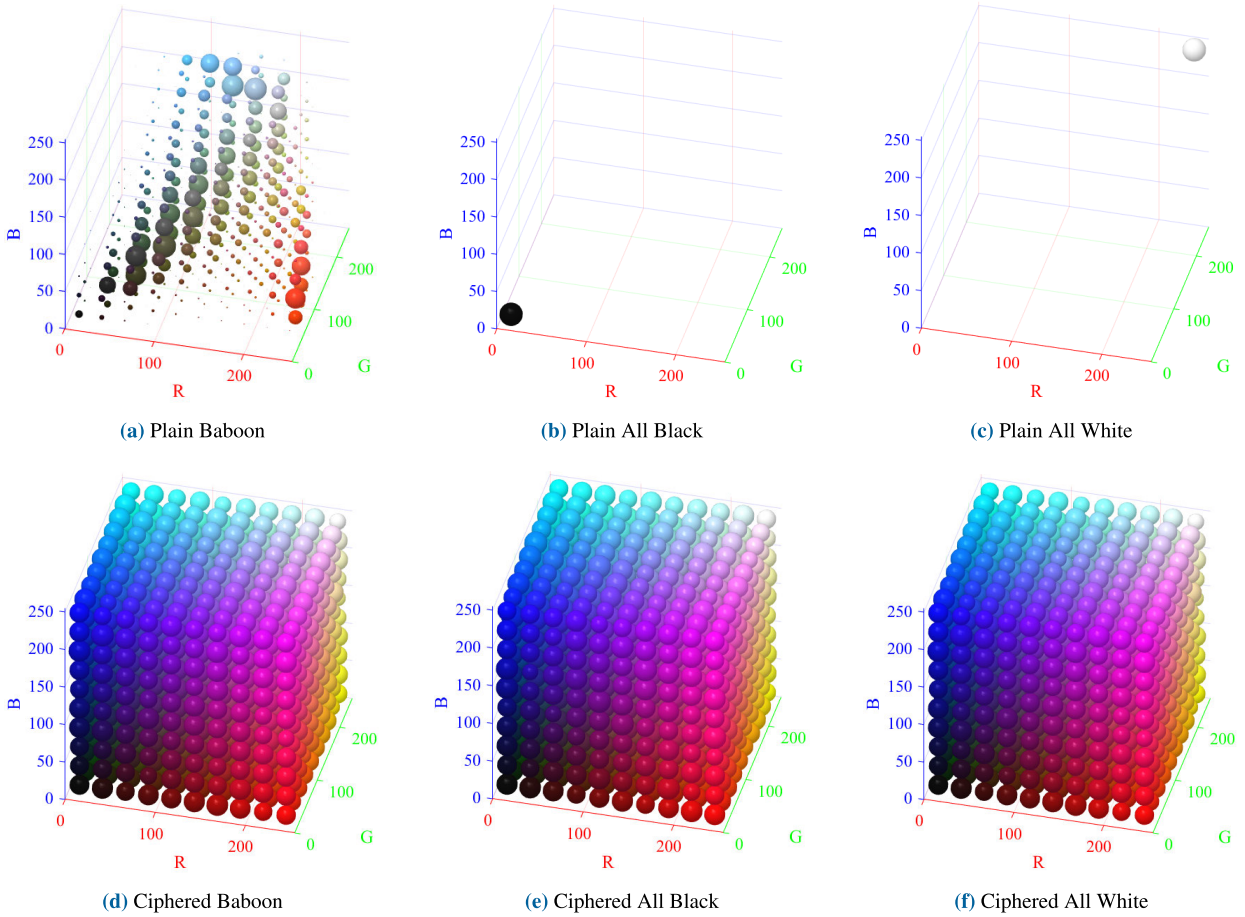
### B. DIFFERENTIAL ANALYSIS

A differential attack is a method to analyze the structure of the encryption algorithm. An attacker observes pixel variations between two similar plain images and their corresponding encrypted images. The differential analysis enables the cryptanalyst to recover the plaintext images without possession of the encryption key is known to be differential analysis [4]. A plain image $P_1$ and 1-bit modified plain image $P_2$ go through the cipher and return images encrypted $E_1$ and $E_2$, respectively. These $E_1$ and $E_2$ must be 100% different to make differential analysis ineffective. Two metrics are employed to observe the resilience of the image encryption algorithm against differential attacks: the Number of Pixel Change Rate (NPCR) and the Unified Average Changing Intensity (UACI). The NPCR determines the number of different pixels at corresponding locations between two encrypted images, $E_1$ and $E_2$, which are generated from plain images differing by only 1-bit. The higher the NPCR value, the greater the difference between $E_1$ and $E_2$ [11]. This metric is calculated using the following Equation (27):

$$N(E_1, E_2) = \sum_{i=1}^{M} \sum_{j=1}^{N} \frac{D(i,j)}{M \times N} \times 100\% \tag{27}$$

In Equation (28), $D(i, j)$ is defines as follow:

$$D(i,j) = \begin{cases} 0, & if\, E_1(i,j) = E_2(i,j) \\ 1, & if\, E_1(i,j) \neq E_2(i,j). \end{cases} \tag{28}$$




(a) Plain Baboon (b) Plain All Black (c) Plain All White

(d) Ciphered Baboon (e) Ciphered All Black (f) Ciphered All White

**FIGURE 18.** 3D Histogram analysis of different images.

**TABLE 12.** Comparative estimate of Entropy.

| Images | Plain Image | | | | Encrypted Image | | | |
|---|---|---|---|---|---|---|---|---|
| | Red | Green | Blue | Mean | Red | Green | Blue | Mean |
| Baboon | 7.7067 | 7.4744 | 7.7522 | 7.6444 | 7.9993 | 7.9993 | 7.9993 | 7.9993 |
| Pepper | 7.3388 | 7.4963 | 7.0583 | 7.2978 | 7.9993 | 7.9992 | 7.9993 | 7.9993 |
| Airplane | 6.7178 | 6.7990 | 6.2138 | 6.5768 | 7.9992 | 7.9993 | 7.9993 | 7.9993 |
| **Mean.** | **7.2544** | **7.2566** | **7.0081** | **7.1730** | **7.9993** | **7.9993** | **7.9993** | **7.9993** |
| Baboon [53] | - | - | - | - | 7.9991 | 7.9981 | 7.9993 | - |
| Airplane [30] | 6.7178 | 6.7990 | 6.2138 | - | 7.9994 | 7.9992 | 7.9993 | - |
| Peppers [9] | 7.3388 | 7.4963 | 7.0583 | 7.2978 | 7.9992 | 7.9993 | 7.9993 | 7.9993 |
| Airplane [54] | - | - | - | - | 7.9992 | 7.9993 | 7.9992 | - |

The other metric is UACI which measure the pixel's intensity at corresponding locations between the encrypted images $E_1$ and $E_2$. The higher value of UACI means both encrypted images are not identical with each other [11]. The UACI is computed using the following Equation (29):

$$U(E_1, E_2) = \sum_{i=1}^{M} \sum_{j=1}^{N} \frac{|E_1(i,j) - E_2(i,j)|}{L \bullet M \times N} \times 100\% \quad (29)$$

In the above Equations (27) and (29), $M \times N$ demonstrates the dimensions of the image, $E_1(i,j)$ and $E_2(i,j)$ represent pixel locations in the *ith* row as well as *jth* column of the evaluated images. The highest possible brightness level a single pixel can achieve in a digital image is represented by $L$.

The theoretical NPCR score for 8-bit image for image size $512 \times 512$ is 99.5893% under significance level $\alpha_{0.05}$. The theoretical UACI scores have defined lower and upper bounds for a given image size under selected significance level $\alpha$. The upper and lower bounds are 33.3730% and 33.5541% for $512 \times 512$ image and significance level $\alpha_{0.05}$ [58], [59]. Table 15, presents the analysis metrics for all tested images, and the proposed system has successfully qualified

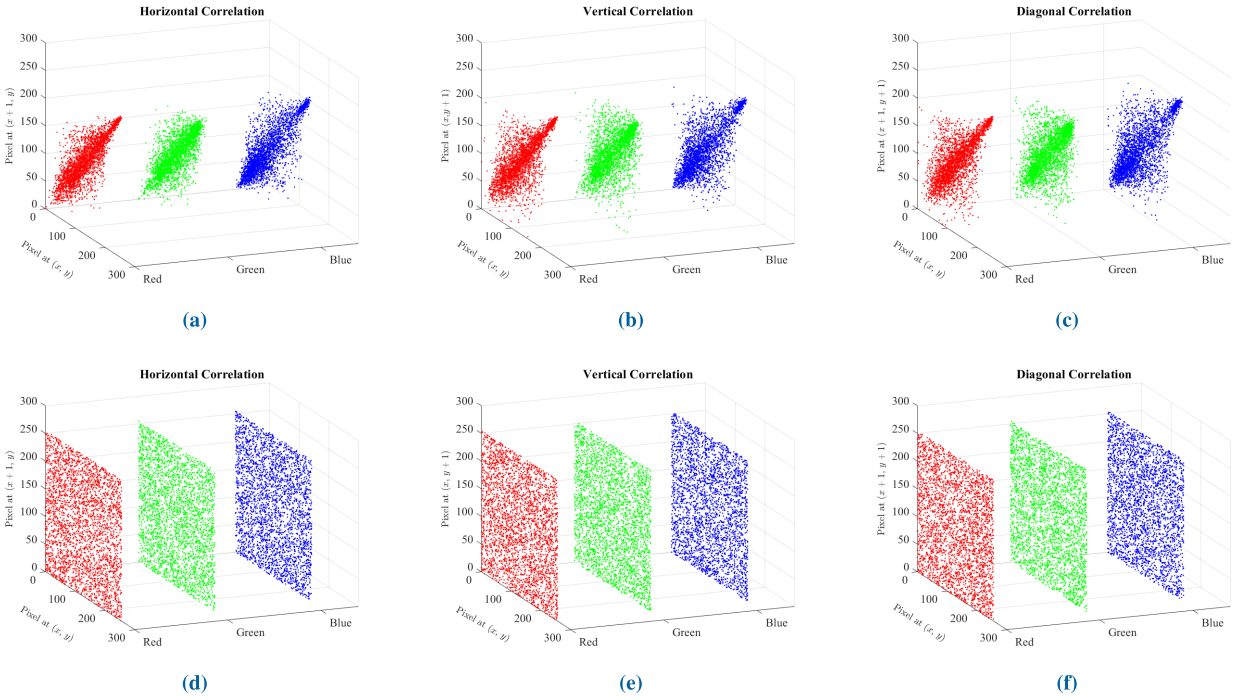



**FIGURE 19.** Correlation analysis: plain Baboon:(a)-(c); encrypted Baboon: (d)-(f).

**TABLE 13.** Analytical score and comparison of LSE.

| Image | Channel | | | |
|---|---|---|---|---|
| | Red | Green | Blue | Mean |
| Baboon | 7.9036 | 7.9026 | 7.9040 | 7.9034 |
| Peppers | 7.9033 | 7.9037 | 7.9032 | 7.9034 |
| Airplane | 7.9021 | 7.9039 | 7.9030 | 7.9030 |
| **Mean.** | **7.9030** | **7.9034** | **7.9034** | **7.9033** |
| Baboon [53] | - | - | - | 7.8949 |
| Baboon [55] | - | - | - | 7.1051 |
| Baboon [26] | 7.9028 | 7.9008 | 7.8996 | 7.9011 |

**TABLE 14.** Chi-Square results of different images with comparison.

| Image | Channel | | | | |
|---|---|---|---|---|---|
| | Red | Green | Blue | Mean | Conclusion |
| Baboon | 250.1094 | 256.6230 | 251.6562 | 252.7962 | Pass |
| Peppers | 275.1309 | 248.5898 | 251.4395 | 258.3867 | Pass |
| Airplane | 279.0742 | 264.1504 | 285.5566 | 276.2604 | Pass |
| **Mean.** | **268.1048** | **256.4544** | **262.8841** | **262.4811** | Pass |
| Peppers [53] | - | - | - | 292.2402 | Passed |
| Baboon [55] | 290.465 | 253.834 | 230.035 | - | Passed |
| Baboon [29] | - | - | - | 256.2598 | Passed |
| Baboon [23] | 274.3828 | 254.7109 | 268.3984 | - | Passed |

both tests under significance level $\alpha_{0.05}$. All the encrypted images have a UACI score above 33.3730% and the NPCR value also qualifies as 99.5893%. From these findings, one can infer that the cipher is robust against differential attacks. Moreover, the encryption method has been compared with other methods found in literature [24], [30], [53], [55], where

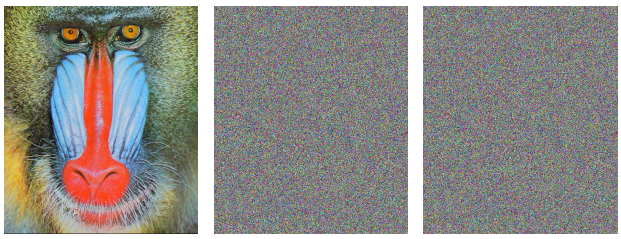



**FIGURE 20.** Key Sensitivity Analysis: (a) Plain Baboon; (b) Encrypted (a) with $\delta_0$ (c) Decrypted (b) with $\delta_1$.

colored images were used for similar assessment as shown in Table 15. Results demonstrate the encryption scheme's resilience against differential attacks.

### C. KEY ANALYSIS

#### 1) KEY SPACE

The essential key space must be appropriate, capable of preventing brute force attacks, and guaranteeing the security of the image cryptosystem. The total addressable key space represents the total quantity of possible keys usable in a cipher. The SHA-256 function is used to transform the secret keys to iterate NEPCS. Since the precision of all the initial seeds $b_0$, $b_1$, $b_2$, $b_3$, $b_4$, $b_5$ and $b_6$ is $10^{-14}$, the key space of the encryption system is $K_{total} = K_{Encryption} \times K_{SHA256} = 10^{171} \cong 2^{581.55}$, which surpass the minimal value of $2^{100} \approx 10^{30}$ [60]. A comparison is represented in Table 16, and it is huge to prohibit brute force attacks to reveal the secret key.

**TABLE 15.** Scores and comparison of differential analysis.

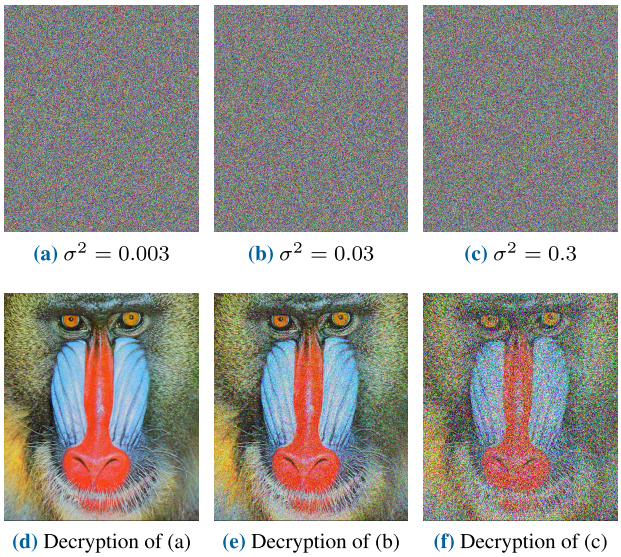
| Image | NPCR (%) | | | | | UACI (%) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Red | Green | Blue | Mean | $\alpha^*_{0.05}$ | Red | Green | Blue | Mean | $\alpha^{*-}_{0.05}/\alpha^{*+}_{0.05}$ |
| Baboon | 99.6090 | 99.6143 | 99.6082 | 99.6105 | Pass | 33.5216 | 33.5048 | 33.4738 | 33.5001 | Pass |
| Peppers | 99.6170 | 99.5960 | 99.6272 | 99.6134 | Pass | 33.4587 | 33.4905 | 33.4501 | 33.4665 | Pass |
| Airplane | 99.6094 | 99.6181 | 99.6265 | 99.6180 | Pass | 33.5226 | 33.4492 | 33.5626 | 33.5115 | Pass |
| **Mean** | **99.6118** | **99.6095** | **99.6206** | **99.6140** | **Pass** | **33.5010** | **33.4815** | **33.4955** | **33.4927** | **Pass** |
| Airplane [24] | - | - | - | 99.6105 | Passed | - | - | - | 33.4632 | Passed |
| Peppers [53] | 99.6151 | 99.6154 | 99.6307 | - | Passed | 33.5246 | 33.3706 | 33.5686 | - | Passed |
| Peppers [55] | 99.6231 | 99.6098 | 99.6002 | - | Passed | 33.5008 | 30.4998 | 33.4634 | - | Passed |
| Airplane [30] | 99.6185 | 99.6125 | 99.6113 | - | Passed | 33.4845 | 33.5153 | 33.5251 | - | Passed |

**TABLE 16.** Comparative analysis of Key space.

| Algorithm | Key Space |
|---|---|
| Proposed | $10^{171}$ |
| Ref. [25] | $10^{112}$ |
| Ref. [31] | $10^{128}$ |
| Ref. [1] | $10^{110}$ |
| Ref. [26] | $2^{448}$ |
| Ref. [27] | $10^{80}$ |

**TABLE 17.** Sensitivity analysis of different keys on images.

| Key Set | Baboon | Peppers | Airplane |
|---|---|---|---|
| $\delta_1(b'_0)$ | 99.6117 | 99.6113 | 99.6114 |
| $\delta_2(b'_1)$ | 99.6162 | 99.6109 | 99.6176 |
| $\delta_3(b'_2)$ | 99.6110 | 99.6053 | 99.5986 |
| $\delta_4(b'_3)$ | 99.6145 | 99.6099 | 99.5960 |
| $\delta_5(b'_4)$ | 99.6016 | 99.6133 | 99.6024 |
| $\delta_6(b'_5)$ | 99.6072 | 99.6096 | 99.6095 |
| $\delta_7(b'_6)$ | 99.6137 | 99.6104 | 99.6057 |
| $\delta_8(\lambda')$ | 99.6112 | 99.6193 | 99.6226 |
| $\delta_9(p')$ | 99.6030 | 99.6100 | 99.6122 |
| **Mean.** | **99.6100** | **99.6111** | **99.6084** |



**(a)** $\sigma^2 = 0.003$ **(b)** $\sigma^2 = 0.03$ **(c)** $\sigma^2 = 0.3$

**(d)** Decryption of (a) **(e)** Decryption of (b) **(f)** Decryption of (c)

**FIGURE 21.** Robustness analysis for Gaussian noise.

### 2) KEY SENSITIVITY

Evaluating a robust encryption system depends on the sensitivity of the secret keys utilized in the encryption and decryption of digital images. For this purpose, the seven user input keys $b_0$, $b_1$, $b_2$, $b_3$, $b_4$, $b_5$, $b_6$, $\lambda$ and prime number $p$ are collectively called key set $\delta_0$ and if one of the $b_i$ or $\lambda$ or $p$ is changed by one bit will create a different key set $\delta_i$. For example, by changing $b_0 = 0.381294613786$ to $b'_0 = 0.381294613787$ in the key set, a new key set called $\delta_1$ will be formed. In this way, nine key sets $\delta_i$ where $i = 1, 2, 3, 4, 5, 6, 7, 8, 9$ are created and used to evaluate the key sensitivity.

Now, the key sensitivity can be measured in two different ways: one is encrypting the plain image with one key set $\delta_i$ and decrypting with another key set $\delta_j$ where $i$ and $j$ are not the same. In this way, the decryption with the wrong key set must fail and the difference must be close to 100%. The plain Baboon image 20(a) is encrypted with $\delta_0$ shown in Figure 20(b) and then 20(b) is decrypted using $\delta_1$ which in turn failed to recover the plain image as shown in Figure 20(a). So, a tiny modification in the key, the cipher is unable to decrypt the baboon, hence the cipher is highly sensitive to minute changes. The Baboon, pepper, and Airplane images are encrypted using $\delta_0$, and then tried to decrypt using $delta_i$ where $i = 1, 2, 3, 4, 5, 6, 7, 8, 9$. The difference is computed as NPCR and the results of baboon, pepper, and airplane images are shown in Table 17.

The other way to analyze the key sensitivity requires a comparison of two encrypted images of Baboon $E_i$ and $E_{i+1}$ where $E_i$ emerges from applying encryption on plain image $P$ using a key set $\delta_0$, while $E_{i+1}$ derives from encrypting $P$ with another key set $\delta_1$. The difference scores get calculated for both images $E_i$ and $E_{i+1}$ and similarly for all key sets mentioned in Table 18. The $\delta_i$ on the top column and 1st row represents the to use to encrypt the plain image $P$ and then the NPCR of two encrypted images with different key sets is computed. The score becomes 0 or zero when the encryption key sets are identical. The key strength of the proposed security system manifests through its 99.60% rating, proving its high tolerance for modifications in secret keys as illustrated in Table 18. Fortunately, for such a

**TABLE 18.** Key sensitivity analysis for Baboon image.

| KeySet | $\delta_0$ | $\delta_1$ | $\delta_2$ | $\delta_3$ | $\delta_4$ | $\delta_5$ | $\delta_6$ | $\delta_7$ | $\delta_8$ | $\delta_9$ |
|---|---|---|---|---|---|---|---|---|---|---|
| $\delta_0$ | 0 | 99.6091 | 99.6023 | 99.6057 | 99.6253 | 99.6035 | 99.6188 | 99.6063 | 99.6168 | 99.6145 |
| $\delta_1$ | - | 0 | 99.6114 | 99.6066 | 99.6068 | 99.6106 | 99.6077 | 99.6106 | 99.6117 | 99.6075 |
| $\delta_2$ | - | - | 0 | 99.6092 | 99.6085 | 99.6159 | 99.6047 | 99.6137 | 99.6099 | 99.6076 |
| $\delta_3$ | - | - | - | 0 | 99.6143 | 99.6136 | 99.5996 | 99.5981 | 99.6157 | 99.6063 |
| $\delta_4$ | - | - | - | - | 0 | 99.6095 | 99.5960 | 99.6025 | 99.6086 | 99.6067 |
| $\delta_5$ | - | - | - | - | - | 0 | 99.6162 | 99.6138 | 99.6190 | 99.6134 |
| $\delta_6$ | - | - | - | - | - | - | 0 | 99.6123 | 99.6222 | 99.6016 |
| $\delta_7$ | - | - | - | - | - | - | - | 0 | 99.6015 | 99.6098 |
| $\delta_8$ | - | - | - | - | - | - | - | - | 0 | 99.6176 |

**TABLE 19.** Quantitative results under Gaussian noise.

| Image | $\sigma^2$ | MSE | PSNR(dB) | NPCR(%) | UACI(%) |
|---|---|---|---|---|---|
| Baboon | 0.003 | 129.25 | 27.01 | 96.48 | 3.55 |
| | 0.03 | 319.45 | 23.08 | 97.76 | 5.59 |
| | 0.3 | 8717.16 | 8.72 | 99.69 | 30.61 |
| Peppers | 0.003 | 125.89 | 27.13 | 94.66 | 3.47 |
| | 0.03 | 310.16 | 23.21 | 95.91 | 5.45 |
| | 0.3 | 8658.57 | 8.75 | 97.79 | 29.55 |
| Airplane | 0.003 | 129.95 | 26.99 | 96.46 | 3.56 |
| | 0.03 | 315.70 | 25.23 | 97.12 | 5.36 |
| | 0.3 | 8576.00 | 8.79 | 99.68 | 29.35 |

**TABLE 20.** Analytical scores of different images using Salt & Pepper noise.

| Image | $\sigma$ | MSE | PSNR(dB) | NPCR(%) | UACI(%) |
|---|---|---|---|---|---|
| Baboon | 0.5% | 96.52 | 28.28 | 0.50 | 0.25 |
| | 5% | 968.69 | 18.26 | 4.98 | 2.49 |
| | 50% | 9681.97 | 8.27 | 49.95 | 24.95 |
| Peppers | 0.5% | 108.05 | 27.79 | 0.50 | 0.25 |
| | 5% | 1052.95 | 17.90 | 4.92 | 2.51 |
| | 50% | 10445.51 | 7.94 | 48.92 | 24.95 |
| Airplane | 0.5% | 105.65 | 27.89 | 0.50 | 0.25 |
| | 5% | 1059.82 | 17.87 | 5.02 | 2.51 |
| | 50% | 10559.28 | 7.89 | 49.98 | 24.97 |

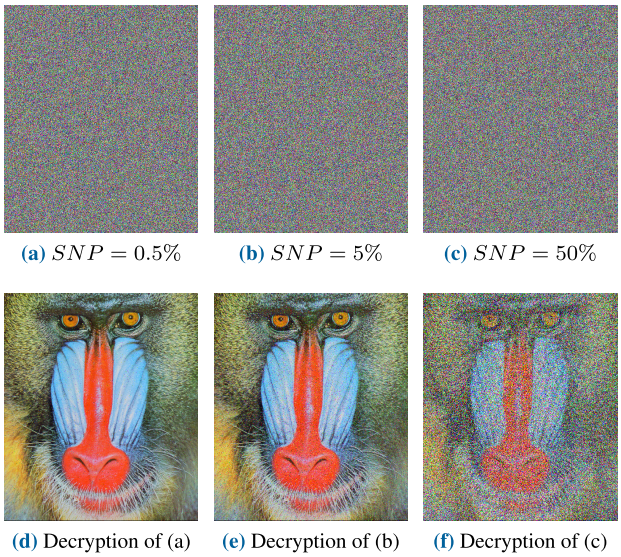
**TABLE 21.** Robustness analysis against occlusion attack.

| Image | Ratio | MSE | PSNR(dB) | NPCR(%) | UACI(%) |
|---|---|---|---|---|---|
| Baboon | 1/16 | 1086.89 | 17.76 | 30.14 | 4.47 |
| | 1/4 | 3914.59 | 12.20 | 79.22 | 15.12 |
| | 1/2 | 6730.28 | 9.85 | 93.85 | 23.73 |
| Peppers | 1/16 | 1277.21 | 17.06 | 30.21 | 4.78 |
| | 1/4 | 4555.66 | 11.54 | 79.22 | 16.13 |
| | 1/2 | 7700.50 | 9.26 | 93.70 | 25.18 |
| Airplane | 1/16 | 1444.44 | 16.53 | 30.26 | 5.02 |
| | 1/4 | 5053.94 | 11.09 | 79.36 | 16.89 |
| | 1/2 | 8467.51 | 8.85 | 93.84 | 26.44 |

chaotic setup, even slight alterations in the key set prove to have unpredictable outcomes. Restoring the original image requires the identification of the keys with utmost precision.

### D. ROBUSTNESS ANALYSIS

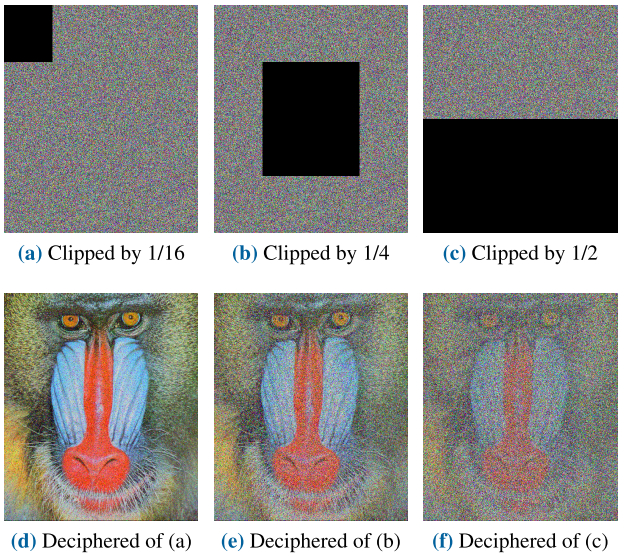
When the encrypted image is transmitted, the information is exposed to interference. A robust algorithm should be able to recover or at least extract the original information embedded in the ciphertext image, even when a part of it is contaminated by interference. The quality of recovered images can be measured using Mean Square error (MSE) and Peak Signal to Noise Ratio (PSNR), which are defined by the following



(a) $SNP = 0.5\%$ (b) $SNP = 5\%$ (c) $SNP = 50\%$

(d) Decryption of (a) (e) Decryption of (b) (f) Decryption of (c)

**FIGURE 22.** Robustness analysis against Salt & pepper noise.



(a) Clipped by 1/16 (b) Clipped by 1/4 (c) Clipped by 1/2

(d) Deciphered of (a) (e) Deciphered of (b) (f) Deciphered of (c)

**FIGURE 23.** Robustness from occlusion attack.

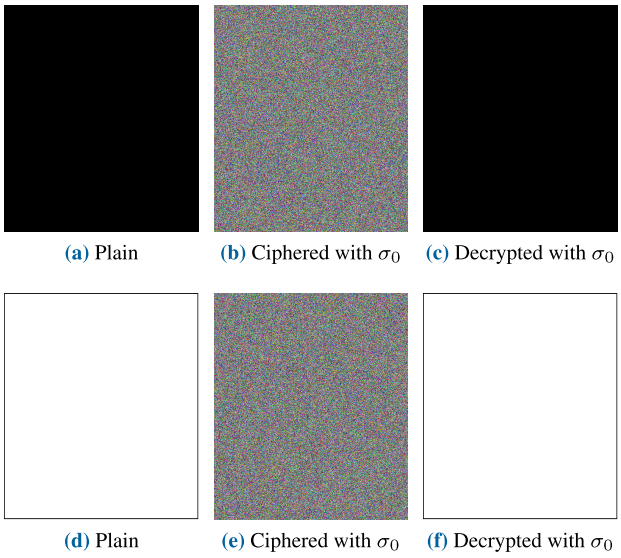



**FIGURE 24.** Visual result of different encryption phases: (a) - (c) All Black Image; (d) - (f) All White Image.

Equations (30) and (31) respectively.

$$MSE = \frac{1}{M \times N} \sum_{i=1}^{M} \sum_{j=1}^{N} \left| A^1(i,j) - A^2(i,j) \right|^2 \tag{30}$$

$$PSNR = 10 \times \log_{10} \left[ \frac{Max^2}{MSE} \right] \tag{31}$$

To evaluate the robustness against communication interference, Gaussian and Salt & Pepper noise uses at different variance levels and occlusion attacks are discussed in the following sections.

#### 1) GAUSSIAN NOISE

A common type of random disturbance called Gaussian Noise introduces images to simulate the real world conditions, where images commonly suffer from some degradation. The results of applying Gaussian noise to determine the performance of the encryption method under various noise levels are presented in Table 19. The robustness of the proposed technique in restoring the quality of an image is evaluated by comparing the MSE, PSNR, NPCR and UACI at different variance levels while keeping the mean value zero. Figure 21 shows encrypted images polluted, and corresponding recovered images at differing levels of Gaussian noise. The recovered images are highly similar to the originals and have successfully undergone noise immunity with image integrity unchanged. Hence, the proposed technique has a quality to resist the Gaussian noise.

#### 2) SALT & PEPPER NOISE

The Salt & Pepper noise (SPN), which could be presented by randomly scattered white, and black pixels, could seriously deteriorate the image's visual appearance and influence the encryption algorithm's efficiency. Typically a cryptographic system is introduced to varying levels of salt and pepper noise, to evaluate it's robustness against SPN. Typically, a cryptographic system is introduced to varying salt and pepper noise levels to evaluate its robustness against SPN. Typically, the performance metrics MSE, PSNR, NPCR, and UACI quantify the system's resilience for salt and pepper noise sources. Researchers have shown these through comprehensive robustness testing, identifying vulnerabilities in the cryptographic image system, and refining the algorithms to be more stable and produce better results in real-world scenarios [11].

The MSE, PSNR, NPCR and UACI values are illustrated in Table 20 for the decrypted images (noise added in encrypted images) and the original plain image. A lower PSNR elaborates the structural dissimilarity between the recovered and original plain images. On the other hand, the higher the MSE score, the higher the distortion or error from the original plain image to the noisy image. The results in Table 20 show that the image encryption system has fair resilience against salt & pepper noise attacks and maintains decrypted image quality.

As Figure 22 shows, even adding salt & pepper noise at different intensities will allow retrieval of the original image. The decrypted images firmly match the original and comparison results show excellent relations between original plain and decrypted images.

#### 3) OCCLUSION ATTACK

An Occlusion attack is an adversarial attack in image processing in which an attacker purposely modifies or distorts some area of an encrypted image or mutants some pixels to conceal some of the information. Such attacks may lead to unauthorized access to sensitive information, falsification of visual content, or total encryption failure in many cases [61]. The results of testing the occlusion robustness of the proposed image encryption algorithm for Baboon, Pepper, and Airplanes at different levels are listed in Table 21. The MSE, PSNR, NPCR, and UACI degrade as the distortion level increases, but the algorithm successfully decrypts the visual information.

The decrypted images with occlusion levels $1/16$, $1/4$ and $1/2$ of the data are shown in Figure 23. Although these images are distorted, remain largely recognizable to the human eye. These results demonstrate that the encryption method is resilient to occlusion attacks up to $1/2$.

### E. CHOSEN PLAINTEXT ATTACK

Mindful opponents exploit encryption vulnerabilities through the use of different images in order to hinder shuffling operations in hopes of obtaining sensitive data. The proposed encryption procedure incorporates multiple shuffling operations, making it resistant to adversary attacks. For testing, we utilized all black and all white images. The output at different encryption scheme phases are depicted in Figure 24, produced impressive results. The plain and encrypted images appeared completely disordered along with smooth histograms, as depicted in Figure 18.

**TABLE 22.** Security analysis of Chosen Plaintext Attack.

| Channel | Entropy | Local Entropy | $\chi^2$ | NPCR(%) | | UACI(%) | |
|---|---|---|---|---|---|---|---|
| | | | | $(P, E_1)$ | $(E_1, E_2)$ | $(P, E_1)$ | $(E_1, E_2)$ |
| Encrypted All Black ($512 \times 512$) | | | | | | | |
| R | 7.9993 | 7.9021 | 243.8672 | 99.5995 | 99.6277 | 49.9405 | 33.4727 |
| G | 7.9994 | 7.9035 | 230.2051 | 99.6361 | 99.6338 | 50.0786 | 33.4517 |
| B | 7.9994 | 7.9031 | 234.1250 | 99.6197 | 99.6098 | 50.0318 | 33.4886 |
| Encrypted All White ($512 \times 512$) | | | | | | | |
| R | 7.9993 | 7.9027 | 249.6094 | 99.6124 | 99.5861 | 49.8973 | 33.4550 |
| G | 7.9994 | 7.9023 | 223.9277 | 99.5968 | 99.5953 | 49.9684 | 33.4757 |
| B | 7.9992 | 7.9030 | 284.1230 | 99.5907 | 99.6067 | 50.0407 | 33.4969 |

The different metrics like entropy, LSE and $\chi^2$, NPCR and UACI are measured for plain ($P$) and encrypted ($E_1$) images as well as two encrypted $E_1$ and $E_2$ (encrypted images obtained from one bit modified plain images). The results are presented in Table 22, which show very impressive scores like entropy ($> 7.9991$), LSE ($> 7.9020$), $\chi^2$ ($< 284.20$), NPCR ($> 99.5860$) and UACI ($> 33.4515$). The security features of this new cryptographic method have been proven successful and resilient toward chosen plaintext attacks.

### F. COMPUTATIONAL COMPLEXITY AND TIME EFFICIENCY ANALYSIS

Each step must be taken into account to evaluate the computational complexity of the proposed encryption algorithm. It consists of six steps. The *key generation* phase begins by hashing the plain image of size $3MN$ using the SHA-256 hash function. The output of the hash function is used to manipulate the keyboard input, therefore, the complexity of the key generation phase is linear that is $\mathcal{O}(M \times N)$. The *initial permutation* permutes the RGB image of size $3 \times M \times N$ employing a sequence of random numbers generated via NEPCS. The permutation involves a sorting algorithm, so the complexity is $\mathcal{O}(M \times N \log(M \times N))$. The *DNA encoding* phase involves linear operations to transform permuted pixels into DNA bases giving the time complexity of $\mathcal{O}(M \times N)$. The *substitution* phase includes zigzag and snake transposition to rearrange the DNA encoded values and Cipher Block Chaining (CBC) encryption that applies DNA XOR and addition using the pseudo-random sequences. All operations in this phase are also linear concerning the image size, leading to a time complexity of $\mathcal{O}(M \times N)$. The *final permutation* is similar in structure to the initial permutation, so the time complexity is $\mathcal{O}(M \times N \log(M \times N))$. Thus, the overall computational complexity of the proposed encryption algorithm can be expressed as $\mathcal{O}(M \times N \log(M \hat{\text{A}} \times \hat{\text{A}} N))$.

Moreover, considering an image encryption system, the focus should also be on the system's efficiency. In order to assess efficiency, how much time is taken by the algorithm to process the plaintext. This article compares time efficiency in Table 23 with various approaches found in literature and finds that our method can process the inputs quickly and efficiently.

**TABLE 23.** Time efficiency of encryption algorithms.

| Algorithm | Encryption Time (second) |
|---|---|
| Proposed | 1.2137 |
| Ref. [28] | 2.0866 |
| Ref. [31] | 3.8206 |
| Ref. [10] | 2.8251 |
| Ref. [27] | 1.33 |

## VII. CONCLUSION

This paper introduces the Normalized Piecewise Exponential Chaotic System, a highly chaotic 1D non-linear system with a broader range of control parameters, high score of Lyapunov Exponent, Sample Entropy, Correlation Dimension as well as resistance ability against the brute force attacks due to remarkable key sensitivity. Further, the NEPCS has successfully qualified for the TestU01 and NIST tests, proving its unpredictability and suitability for the cryptographic applications. Moreover, the efficacy of NPECS is proved via DNA-based cryptographic algorithm, which uses the initial permutation to shuffle the pixels and the final permutation to shuffle the DNA bases of each pixel. The CBC substitution is packed within key less transpositions of Zigzag and Snake patterns. The experimental results prove the algorithm's security against statistical and differential attacks as well as its effectiveness for the industrialÂ applications.

The only limitation of the proposed map is its exponential complexity which minutely takes more time as compared to the Logistic, PWLCM and enhanced Logistic map. In future, we aim to optimize the computational complexity of NEPCS by simplifying its structure to reduce the exponential complexity $O(e^x)$. Although the MATLAB implementation has constant-time complexity for the exponential function, we are still focused to extend its efficiency for other platforms. Currently, the proposed encryption algorithm encrypts 8-bit color images using RGB model. In future we intend to broaden its application for 10-bit color images.

## ACKNOWLEDGMENT

The authors would like to thank the School of Technology and Innovations, VTT-Technical Research Centre of Finland Ltd., Espoo, Finland, and the Institute of Mathematics, Khwaja

Fareed University of Engineering, Science and Technology, Rahim Yar Khan, Pakistan, for supporting this study. They are also thankful for the support of the Artificial Intelligence and Data Analytics Laboratory (AIDA), CCIS, Prince Sultan University, Riyadh, Saudi Arabia.

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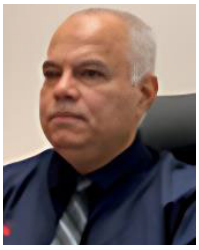
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